Store-Entry Coupons: A Large-Scale Field Experiment

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Abstract

New technologies, such as in-store kiosks and mobile apps, are transforming grocery coupons by enabling delivery when customers are entering the store. We investigate the effectiveness of these "bottom-of-the-funnel" promotions using a large-scale field experiment across 71 product categories at 147 stores. The retailer in our study distributes *store-entry coupons*: printed discounts delivered via kiosks at the store entrance and redeemable on the current trip. On average, store-entry coupons cause a nearly fivefold increase in sales of the promoted brands and drive substantial category expansion. The category expansion is not just temporal substitution. Instead, store-entry coupons appear to be effective at inducing store switching: customers purchase at the focal retailer immediately with a certain discount, rather than purchasing at another retailer in the future, where the availability of a discount is uncertain. Cross-retailer substitution makes store-entry coupons less valuable for manufacturers, but especially valuable for retailers.

Keywords: store-entry coupons, in-store retail media, coupons, field experiments, causal effects, incrementality

1. Introduction

New technologies are transforming the way retailers promote consumer packaged goods (CPG). Retailers such as CVS, Winn-Dixie, Food Lion, and Stop & Shop now distribute coupons via in-store kiosks (Hamstra 2023). Kroger, Target, Publix, and Best Buy have introduced mobile apps that consumers can use to browse coupons in-store (Berthiaume 2022). Other emerging technologies include digital in-store signage and intelligent shopping carts (Grewal et al. 2023). These technologies are disrupting the $430 billion CPG coupon industry by delivering promotions to customers at the bottom of the purchasing funnel, when they are about to make a purchase.

Whether these "bottom-of-the-funnel" promotions are effective in physical retail remains an open question. On the one hand, distributing coupons in-store reduces the gap between promotional exposure and the purchase decision. Amazon's sponsored search and on-site advertising share this characteristic, and the effectiveness of these lower-funnel interventions has helped Amazon become the world's third-largest advertising company (Statista 2026). On the other hand, because in-store promotions target customers who have already committed to a store visit, they may merely subsidize existing purchase intent. If shoppers simply apply the discounts to planned purchases or use them to pull demand forward from future weeks, the coupons will not generate incremental growth. Determining which of these arguments prevails is an important empirical question.

We investigate the effectiveness of these promotions using a large-scale field experiment at a major supermarket chain in Germany. We study *store-entry coupons*: coupons delivered when customers scan their loyalty cards at kiosks at the store entrance, before shopping begins. Store-entry coupons are the mirror image of checkout coupons, which are printed at the register after purchase for redemption on a future trip. Store-entry coupons are delivered before the basket is filled and can be redeemed on the current trip. They are an example of in-store retail media (advertising and promotions delivered through touchpoints inside physical stores), and our study provides the first large-scale experimental evidence on this emerging channel.

The experiment was conducted in a business-as-usual environment by distributing random coupons using the retailer's couponing system. The random coupon assignments exogenously varied the promoted brands and discount depth across different customers. We use the retailer's loyalty program to link coupon exposures to consumers' individual purchases and report the incremental

(causal) effects across 101 CPG campaigns spanning 71 categories.

Store-entry coupons dramatically increase purchases of promoted brands at the focal retailer. On average, receiving a store-entry coupon causes purchase incidence to increase from 0.33% to 1.61% on the current visit. A logical concern is that these unit gains do not translate into revenue because discounts offset the additional sales. Our findings assuage this concern. The average revenue increased from €6.59 to €34.09 per 1,000 coupons (per mille), with no statistically significant revenue loss in any of the 101 campaigns, despite an average discount of roughly 24%.

These incremental brand sales do not come at the expense of competing brands within the same category. On average, the category-level revenue effect (€31.03 per mille) is as large as the brand-level effect (€27.50 per mille), so store-entry coupons expand the category, rather than merely shift demand within it. The category expansion does not simply represent temporal substitution either: when averaging across brands, we find no evidence of a post-promotion dip or stockpiling.

We argue that an important source of category expansion is store switching. Grocery shoppers routinely split purchases across several retailers, so a store-entry coupon received at the start of a shopping trip can pull into the current basket purchases they would otherwise have made at a competing retailer on a future trip. The logic is clearest in categories with fixed consumption, such as detergent or toilet paper, where a coupon cannot induce a household to consume more, so any incremental sale must be diverted from another purchase occasion. For these categories, we observe category expansion even if customers had not purchased the category from the focal retailer in the previous three months, suggesting they had been buying the category elsewhere.

Two additional empirical findings support store switching. First, we compare focal stores located near and far from competitors. Both groups show immediate category expansion, but over the following five weeks, the effect persists in stores near competitors while fading in more isolated stores. This pattern is consistent with store switching: when a competitor is nearby, the coupon can divert a purchase the customer would otherwise have made there, leaving the focal retailer with a persistent gain. In contrast, in more isolated stores, the incremental purchase is more likely pulled forward from the customer's own future visits to the focal retailer.

Second, we use household panel data that span the retailer's staged introduction of the

coupon system, to estimate the effect of the store-entry coupons. The introduction raised the focal retailer's sales by about 8%, while lowering the main competitor's by about 4%. Moreover, across the nine merchandising groups, we observe a positive relationship between the category expansion in the panel data and the treatment effects in our experiments.

A potential concern is that the effects operate through the retailer's targeting algorithm rather than through the coupons themselves: an initial coupon could trigger additional targeted coupons for the same brand on later visits. We test this channel directly. The treatment effects materialize on the day the coupon is received, before any algorithmic response is possible, and subsequent focal-brand coupon delivery is essentially identical for treated and control customers across the campaigns in our study. The effects, therefore, reflect customer responses to the coupons, rather than differences in downstream coupon targeting.

Our findings have opposite implications for manufacturers and retailers. For manufacturers, store switching undermines the benefits of the positive brand effects in the focal store. For retailers, category expansion is favorable, particularly given the evidence that temporal substitution plays only a limited role. A marketing promotion that expands category sales by stealing sales from competitors is a very attractive tool for a retailer.

Our study provides causal evidence of what firms can expect when adopting store-entry coupons. However, we caution that these data describe customer behavior at a single supermarket chain and rely on a specific distribution mechanism: coupons delivered at the store entrance, before customers reach the aisle. While this timing allows us to capture "bottom-of-the-funnel" effects, additional research is required to further study generalizability across different retail settings.

Outline of the Paper

The paper proceeds with a review of the related research in Section 2. We describe the data and experimental setting in Section 3 and discuss the identification strategy in Section 4. We document outcomes for the promoted brand in Section 5, and investigate category expansion in Section 6. The paper concludes in Section 7.

2. Related Literature

There is a rich literature studying how consumers respond to coupons and price promotions. We first review the literature on in-store promotions and then discuss research related to category expansion.

2.1 In-Store Promotions

Previous investigations of in-store promotions include coupons distributed at the shelf, such as coupons dispensed from J-hooks located directly in front of the promoted product (Dhar and Hoch 1996) or peel-off coupons placed on individual packages (Raju et al. 1994). Dhar and Hoch (1996) conducted two field experiments at 86 Chicago supermarkets. They found that on-shelf coupons yielded sales increases similar to price discounts, but because only 55% of purchasers redeemed coupons, the revenue and profit increases were much larger for the coupons. Raju et al. (1994) conducted a series of quasi-experiments at a UCLA student store in which they placed coupons on 14-oz and 20-oz soda-fountain drinks. They conclude that on-pack coupons lead to larger market share and category sales lifts than peel-off and in-pack coupons.

The store-entry coupons in our study are distributed when customers start their shopping trips. In contrast, on-shelf and peel-off coupons arrive in the middle or at the end of the trip. This distinction is important. Coupons distributed at the start of a trip can affect whether a customer visits an aisle, while this is less likely to occur with on-shelf or peel-off coupons. We can also contrast these coupons with coupons delivered at home (Bawa and Shoemaker 1987; Venkatesan and Farris 2012). At-home coupons are received earlier in the purchase funnel and can affect the frequency and choice of which store to visit, whereas store-entry coupons cannot. Store-entry couponing is a growing component of in-store retail media. Retailers increasingly deliver advertising and promotions through digital touchpoints inside their stores, including entrance kiosks, retailer apps, and digital signage (Gabel et al. 2024; Herhausen et al. 2026).

Two studies consider coupons delivered as customers enter a store. Heilman et al. (2002) distributed printed coupons to 105 customers intercepted at the entrance of two St. Louis grocery stores. In a similar study, Hui et al. (2013) distributed printed coupons to 90 customers entering a grocery store in Pittsburgh. Both studies document the impact of the coupons on "unplanned"

purchases (in categories not on the shopping list). The interactions between researchers and customers introduce a potential limitation to both studies, by introducing a risk that the interactions themselves could impact customers' behavior in the store (Pollay 1968).[1]

Our paper contributes to this literature by investigating the effectiveness of store-entry coupons using a large-scale field experiment. The store-entry coupons we study are a distinctive instrument: unlike the shelf, peel-off, and at-home coupons studied previously, they are delivered after the customer has chosen the store but before the basket is filled. Conducted in a business-as-usual retail environment, the study unobtrusively embeds the intervention into shoppers' purchase journeys without making participants aware of the experimental manipulation. The loyalty program data help us link the coupon distribution to purchase outcomes and evaluate incremental sales by different customer segments. The findings demonstrate that store-entry coupons are effective for driving sales of the promoted brands and category expansion at the focal retailer.

2.2 Category Expansion

Beyond the effects on the promoted brand, prior research has studied whether coupons can contribute to category expansion by increasing consumption or inducing stockpiling. Chiang (1995) finds that for goods with fixed consumption rates, such as laundry detergent, promotions do not drive category growth. In contrast, Ailawadi and Neslin (1998) argue that for other categories like yogurt, increased household inventory can accelerate consumption rates. Nijs et al. (2001) further observe that price promotions can generate category expansion that persists for up to ten weeks (see also Dekimpe et al. 1998; Mela et al. 1998).

The existing literature on in-store promotions offers limited insight into these category-level dynamics. Heilman et al. (2002) and Hui et al. (2013) both focus on customer basket sizes rather than total category sales. In the Dhar and Hoch (1996) study, the effects on category expansion are confounded by concurrent promotional activity on competing brands during the study period. Raju et al. (1994) also explicitly acknowledge that they do not investigate category expansion.

One explanation for category expansion is that coupons can increase consumption via impulse

[1] In addition, Chiou-Wei and Inman (2008) study a setting in which "consumers browse the Web at home for any suitable coupons available on the store coupon pages and print them out. In the case of forgetting and/or misplacing, they can also get access to the coupon pages by running their shopping cards through terminals at the store and printing out the coupons" (p. 297). The browsing and printing at home is very similar to clip-at-home coupons.

purchasing. A key motivation for retailers' in-store marketing activities, including store-entry coupons, is to increase impulse purchasing (Abratt and Goodey 1990; Zhou and Wong 2004). Impulse purchases are unplanned (see the discussion in Shapiro 2001), and often include a "powerful and persistent urge to buy something immediately" (Rook 1987, p. 191). Bucklin and Lattin (1991) propose a two-state model of purchase incidence and brand choice in which consumers shop in two ways. Shoppers who have planned their shopping trips are unresponsive to in-store promotions. In contrast, shoppers who have not planned their trips make impulsive decisions in the store and respond more strongly to in-store promotions.

Alternatively, retailers are often concerned that in-store coupons merely subsidize purchases that would have occurred anyway, shifting future demand to the present. Literature on price promotions frequently supports this "temporal substitution" hypothesis, documenting a post-promotion dip in sales following a promotion (Gupta 1988; Chiang 1991; Grover and Srinivasan 1992; Bell et al. 1999). Household stockpiling is often cited as the driver of this dip, though empirical evidence linking a product's "suitability for stockpiling" to promotional response remains mixed. Some studies report a positive relationship (Narasimhan et al. 1996; Bell et al. 1999; Ailawadi et al. 2006; Fok et al. 2006), whereas others report a negative or non-significant relationship (Nijs et al. 2001; Lim et al. 2005; Osuna et al. 2016).

Our study systematically investigates how store-entry coupons impact sales of both the promoted brand and overall category sales. The scale and scope of the study allow us to document the effects across customers and product categories, providing evidence that the observed category expansion is not merely temporal substitution. Instead, using two different datasets, we provide convergent evidence consistent with store switching. In the next section, we introduce our empirical setting and discuss the design and implementation of the field experiment.

3. Data Overview

The data for our study were provided by a large German grocery retailer. The stores use a traditional supermarket format, and during the experiment, the retailer operated over 500 stores. The product range included over 30,000 SKUs spanning food, beverages, and non-food products, such as laundry detergents and paper towels. The stores use a Hi-Lo price format, and the average

price point is similar to that of its moderately priced competitors. The retailer maintains a loyalty program to collect customer data and distribute store-entry coupons. Our dataset contains the purchases of all customers who shopped in the retailer's 147 stores in one of Germany's federal states, including over 250,000 customers in the retailer's loyalty program.[2]

The retailer communicates with loyalty program members through self-service kiosks (see Figure 1). The kiosks allow customers to print store-entry coupons and check their loyalty point balances. During the study, the kiosk system was the only touchpoint between the retailer and its loyalty program customers.

Figure 1: In-Store Kiosk and Coupon Printout





Notes: The figure depicts the self-service kiosk (left) and the coupon printout (right). The printout contains seven coupons valid for the same day, regular and discounted prices, and the discount rate in percent.

Customers typically interact with the kiosks and obtain coupons when entering the store. After customers present their loyalty cards, the kiosk prints seven coupons, each of which offers a discount on a single brand. When customers show their loyalty card at the checkout, they automatically receive these discounts if they purchase any of the brands represented by the coupons. If customers use the kiosk system multiple times in one day, they receive the same coupons; the

[2]Loyalty program participants account for 21.1% of the retailer's total revenue. The penetration tends to be larger in suburban residential areas with lower household incomes (up to 43.7%).

coupon assignments are determined the night before and are valid for a day.

The retailer manages coupons in campaigns, where each campaign promotes a specific brand's products to loyalty program members over a set period. The campaigns are part of the retailer's retail media offering: the brands initiate the campaigns, set the available discount levels, and fund the discounts, which is standard practice in the industry. In our analysis, we use these campaigns as our primary unit of observation to report how coupons affect sales of the promoted products. We assigned each campaign to a product category using the retailer's internal category map. These category definitions are relatively narrow. Examples include frozen pizza, cat food, deodorant, and juice. On average, a category includes 103 SKUs, and the promoted brands in our experiment represent 13.3% of the revenue in their corresponding categories (see Web Appendix A for a complete list).

The retailer allocates coupons to customers based on their past spending. For example, the targeting system prioritizes coupons for the brands that a customer has purchased before. We will label coupons that the system chooses to distribute to individual customers as *targeted coupons*. The system ensures customers' daily coupons do not include multiple targeted coupons for the same or competing brands (in the same category).

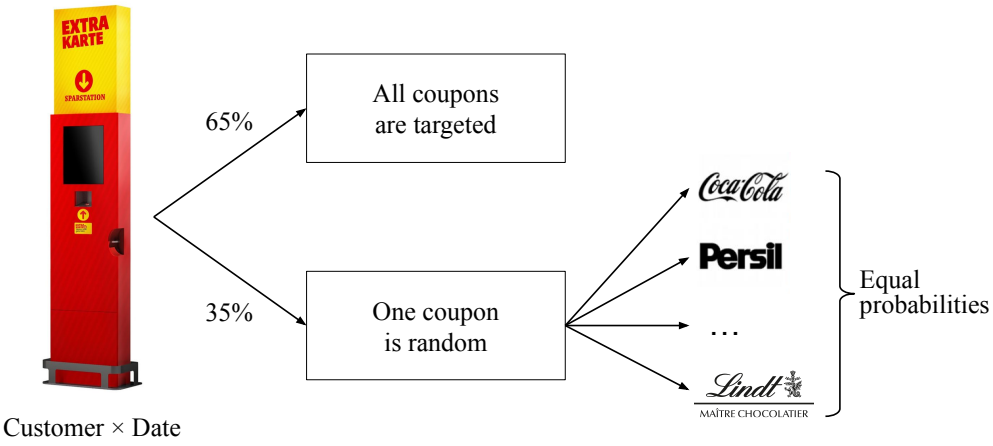
3.1 Experimental Design

We obtained data from an experiment that we helped design and implement for the retailer, and which became part of the retailer's standard practice. The experiment distributed random coupons to both provide training data and evaluate the performance of the couponing system.

We illustrate the experimental design in Figure 2. By default, all seven coupons are targeted. In the experiment, in 35% of kiosk visits, the couponing system replaces one of the targeted coupons with a random coupon. The random coupons feature a randomly selected brand with a random discount depth ranging from 5% to 50%. The coupon assignments and randomization are implemented daily so that, on a given day, each customer has a 65% chance of receiving only targeted coupons, and a 35% chance of receiving a random coupon.

If a customer receives a random coupon, the random coupon replaces one of the seven targeted coupons. The targeted coupon that is replaced is randomly selected. The only exception is if the

Figure 2: Experimental Design with Random Coupons



printout already includes a targeted coupon for the same brand as the random coupon, the random coupon replaces this targeted coupon (ensuring that the printout never includes two coupons for the same brand).

We emphasize that the experiment only includes customers enrolled in this retailer's loyalty program, so our results are limited to this group. The retailer price discriminates by restricting coupon distribution to loyalty program members, deliberately excluding customers who have opted not to join the program from receiving any coupons.[3]

The retailer's couponing system records which coupons are distributed to customers, including the featured brands and discounts. The system also logs which coupons are targeted and which coupons are random (if any). However, the retailer does not record which targeted coupons are replaced by the random coupons, meaning there is no counterfactual policy logging in the system (Johnson et al. 2017).

3.2 Descriptive Statistics

Table 1 reports descriptive statistics. The dataset includes 1,503,000 kiosk visits across 158 different dates (November 2015 to May 2016). Our unit of observation is a customer×date combination with at least one kiosk visit. We focus on 88,840 random coupons distributed over 101 CPG campaigns that had regular sales prior to the experiment, distributed at least 400 random coupons, and had

[3]One may be concerned that loyalty program members are store-loyal customers who would purchase the promoted products anyway. However, household panel data suggest otherwise: members routinely purchase groceries at many competing retail chains. We report this evidence in Web Appendix I.

no eligibility criteria.[4] We provide additional details about sample construction in Appendix A and randomization checks in Web Appendix B. The final sample spans 71 product categories, including food, drinks, and non-food CPG categories. The average discount per random coupon was 23.95%, and the redemption rate for random coupons was 1.61%.

Table 1: Experimental Data Descriptive Statistics

Number of campaigns	101
Number of categories	71
Number of random coupons	88,840
Number of shopping trips with kiosk usage	1,503,000
Average discount per random coupon	23.95%
Redemption rate for random coupons	1.61%
Average face value per redeemed random coupon	€0.84

Our analysis focuses on incremental changes in purchase incidence, quantity, and revenue (see Table 2). Brand managers who execute and evaluate campaigns typically focus on quantity and revenue, and the retailer reports these outcomes in its campaign dashboard. The revenue measure offers the advantage of accounting for price discounts. We consider incidence because we can measure treatment effects for this binary variable with high precision. Comparing multiple metrics also allows us to evaluate the robustness and consistency of our findings.

Table 2: Purchase Outcomes

Metrics	**Description**	**Type**
Purchase Incidence	Did a customer purchase or not?	$\{0, 1\}$
Quantity	How many units did a customer purchase?	$\mathbb{N}^0$
Revenue	How much did the customer spend?	$\mathbb{R}^0$

Our primary analysis in Section 5 focuses on the day that the customer used the kiosk to obtain store-entry coupons; in Section 6, we also extend the measurement window to include purchases in subsequent weeks. We next describe an identification approach that estimates the

[4]Some brand managers restricted the distribution of coupons to a subset of customers (e.g., customers who had recently purchased in the category). To avoid selection effects and facilitate our comparison of treatment effects, we excluded brands that used such eligibility criteria.

average treatment effect for each coupon campaign.

4. Identification Approach

Our goal is to measure the average treatment effect (ATE) of store-entry coupons for loyalty program customers on days that they use the kiosk (customer×date observations). The ATE captures the average impact of providing a coupon for the focal brand, compared to not providing a coupon for the focal brand:

$$\text{ATE} \equiv \mathbb{E}[Y(C=1) - Y(C=0)], \tag{1}$$

where $Y(C=1)$ and $Y(C=0)$ indicate potential outcomes with and without a coupon, and the expectation integrates over customer×date observations. Y refers to the outcome metrics described in Table 2.

4.1 Estimator

To estimate Equation (1), we can interpret the random sample of customers who received a random coupon for the focal brand as a "treatment" sample. Outcomes for these customers directly provide an estimate of $\mathbb{E}[Y(C=1)]$. The remaining observations are a random draw from the same customer population as the treatment sample, and they did not receive a *random* coupon for the focal brand. However, we cannot use their average outcomes to directly estimate $\mathbb{E}[Y(C=0)]$, because some of these customers received a *targeted* coupon for the focal brand.

To account for the focal brand's targeted coupons, our identification approach leverages random coupons for non-focal brands to infer the ATE for the focal brand. We summarize our identification approach here and provide additional details in Appendix B. For each brand, we define three groups of customer×date observations:

1. *Focal*: Observations in which a customer received a random coupon for the focal brand. This is the treatment sample.
2. *NonFocal*: Observations in which a customer received a random coupon for any non-focal brand.
3. *NoRandom*: Observations in which a customer did not receive any random coupons (all coupons are targeted).

The three groups are random draws from the same customer population, so that differences in their outcomes reflect the coupon assignments rather than differences in customer characteristics. The groups are mutually exclusive and representative of the overall customer×date population. The *Focal* observations always received a coupon for the focal brand, due to the random assignment. The *NonFocal* and *NoRandom* observations did not receive a random coupon for the focal brand in the experiment, but they could receive a targeted coupon for the focal brand.

We estimate the average treatment effect for the focal brand using the following equation:

$$\widehat{\text{ATE}} = \underbrace{\bar{Y}(\text{Focal})}_{\mathbb{E}[Y(C=1)]} - \underbrace{\left[\bar{Y}(\text{NoRandom}) - 1/p \cdot \left(\bar{Y}(\text{NoRandom}) - \bar{Y}(\text{NonFocal})\right)\right]}_{\mathbb{E}[Y(C=0)]} \tag{2}$$

where $\bar{Y}(\cdot)$ indicates the average outcome in the corresponding group, and p is the probability that a random *non-focal* coupon replaces a targeted *focal* coupon. This probability is defined over observations in which the customer is assigned to receive both a random non-focal coupon and a targeted focal coupon. In our setting, there are seven targeted coupons that could be replaced by the non-focal coupon, so there is a one in seven chance that the targeted coupon for the focal brand is the one that is replaced ($p = 1/7$).

The treatment effect estimator in Equation (2) contains two components. The first component captures the average outcome in the treatment sample, *Focal*, to estimate $\mathbb{E}[Y(C = 1)]$. The second component estimates $\mathbb{E}[Y(C = 0)]$ using the outcomes in the *NoRandom* and *NonFocal* conditions. Intuitively, the experimental design ensures the equivalence of observations in these two conditions. However, in *NonFocal*, random non-focal coupons replace a proportion p of the targeted coupons, so there are fewer focal targeted coupons in *NonFocal* than in *NoRandom*. If this is the only cause of the different outcomes between the two conditions, we can recover the effect of the targeted coupons in *NoRandom* by scaling the difference in outcomes between *NoRandom* and *NonFocal* by $1/p$. We obtain $\mathbb{E}[Y(C = 0)]$ by subtracting this effect from $\bar{Y}(\text{NoRandom})$. We illustrate this intuition using an example in Appendix B.

Empirically, the adjustment for targeted coupons has little impact on the findings in our

study. The simple difference-in-means estimator is

$$\widehat{\text{ATE}}_{\text{DiM}} = \bar{Y}(\text{Focal}) - \bar{Y}(\text{NoRandom}), \tag{3}$$

and comparing it with Equation (2) yields the exact relationship

$$\widehat{\text{ATE}} = \widehat{\text{ATE}}_{\text{DiM}} + 1/p \cdot (\bar{Y}(\text{NoRandom}) - \bar{Y}(\text{NonFocal})). \tag{4}$$

The two estimators coincide when $\bar{Y}(\text{NoRandom}) = \bar{Y}(\text{NonFocal})$. This can occur when there are no targeted coupons for the focal brand, or when targeted coupons for the focal brand have a zero treatment effect. Empirically, the campaign-level estimates from the two estimators are highly correlated (0.92–0.96 across the incidence, quantity, and revenue outcomes), and the difference-in-means estimates are smaller in absolute magnitude by roughly one-fifth of the $\widehat{\text{ATE}}$.[5]

We can consider how the features of our experimental design and estimation compare with other field experiments in the literature. We make two observations. First, we use the firm's standard system to distribute store-entry coupons. In contrast, Heilman et al. (2002) study coupon effectiveness by intercepting customers as they entered the store, which was not part of normal store operations. There are benefits of operating within a business-as-usual environment: scale is more easily achieved and representativeness is enhanced when marketing actions are reflective of actual firm practice. In our case, we were able to experiment across many brands with thousands of coupon treatments using the same timing and format that the firm uses in production.

Second, because our experiment is implemented as part of normal firm operations, business constraints can interfere with causal inference. In our case, targeted coupons from the coupon system can contaminate the control condition. Similarly, in digital advertising, the targeting algorithms from the advertising platforms can complicate causal inference (Johnson 2023). Our solution to causal inference shares features of the placebo ad (Hoban and Bucklin 2015) and ghost ad (Johnson et al. 2017) approaches that have been used in the digital advertising literature. Akin to placebo ads, we use non-focal coupons for identification because we do not expect them to impact

[5] For most brands, only a very small share of observations in *NoRandom* and *NonFocal* conditions received a targeted focal coupon, while *all* observations in the *Focal* condition received a (random) focal coupon. The difference between $\widehat{\text{ATE}}$ and $\widehat{\text{ATE}}_{\text{DiM}}$ is therefore negligible in most cases.

customer behavior for the focal brand. Like ghost ads, we adjust our estimates of the outcome in the control condition to account for the effects of the firm's targeting algorithm.

The absence of a conventional holdout was a deliberate design choice. The experiment ran inside the retailer's production couponing system, in parallel with live campaigns booked by the brands, which the retailer could not pause. The retailer instead randomly allocated 5% of all coupons (35% of kiosk visits receive one random coupon out of seven). This single randomization layer served two purposes: the exogenous variation trained the recommender system, and the random coupon mechanically displaces a targeted one, so the withholding of targeted coupons occurs within the same layer. We leverage these platform-specific design features in our identification approach.

4.2 Identification Assumptions

The proposed estimator relies on two assumptions commonly used in the causal inference literature (Angrist et al. 1996; Johnson et al. 2017). First, we make the stable unit treatment value assumption (SUTVA). SUTVA requires that the potential outcomes for any given observation do not vary with the treatment assignment for other observations. In our context, this implies that receiving a coupon does not affect the shopping behavior of other customers in the store at the same time, and receiving a coupon does not change the potential outcomes for that same customer on subsequent dates. Second, we assume that the coupon treatment effects do not extend across brands. This exclusion restriction implies that replacing a focal coupon with a random non-focal coupon shifts a potential outcome from $Y_i(1)$ to $Y_i(0)$. In Appendix B, we show that the proposed estimator is unbiased under these two assumptions and we derive its variance.

In principle, the panel structure of our data and competitive spillovers can introduce threats to these assumptions. Regarding SUTVA, we note that the absence of *cross-customer* spillovers is plausible, but using multiple customer×date observations per customer introduces the possibility of temporal spillovers (such as forward-buying), altering a customer's subsequent potential outcomes. Regarding the exclusion restriction, we recognize that competing brands' coupons could affect sales of the focal brand. Our identification approach naturally partials out these competitive effects in the "control" condition, but targeted competitive coupons can co-occur with random focal coupons, which could introduce bias.

We explicitly address both concerns in Web Appendix C by replicating our analysis with a restricted sample. For temporal spillovers, we isolate only the first customer×date observation for every customer in the experiment, and for competitive spillovers, we restrict the sample to campaigns without competitor coupons. We caution that these are robustness checks rather than a full identification solution: the data still reflect repeated customer behavior and store-level shopping routines.

Reassuringly, the campaign-level point estimates are essentially unchanged in the restricted sample, and the campaign-level estimates correlate at approximately 0.80 across the two samples.[6] This robustness is expected in our setting: competitive coupons are rare (co-occurring in $< 1\%$ of "treatment" observations), and their spillover effects are negligible compared to the main focal effects. While temporal spillovers are harder to dismiss *a priori*, the evidence from the first-observation sample suggests that potential temporal dependencies do not affect our conclusions.

4.3 OLS Implementation

We implement the proposed identification approach using an OLS regression framework. Specifically, we stack the observations from the *NoRandom*, *NonFocal*, and *Focal* groups, and then estimate the following regression model:

$$Y_i = \alpha + \beta_F \cdot \text{Focal}_i + \beta_{NF} \cdot \text{NonFocal}_i + \boldsymbol{\gamma} \cdot \text{Controls}_i + \varepsilon_i, \tag{5}$$

where Focal_i and NonFocal_i are indicator variables for the respective experimental groups, and Controls_i represents a vector of covariates. We then calculate the estimated average treatment effect for the focal brand by combining the estimated coefficients:

$$\widehat{\text{ATE}} = \hat{\beta}_F - \frac{1}{p} \cdot \hat{\beta}_{NF}. \tag{6}$$

Without controls, this regression-based approach is mathematically equivalent to the estimator in Equation (2). The baseline group is *NoRandom*, so the intercept α represents the average

[6] The share of statistically significant campaign-level effects is lower in the restricted sample (30–36% versus 56–66%). This is a mechanical consequence of the smaller sample. The campaign-level standard errors inflate by a median factor of 1.5, in line with the factor of 1.7 implied by the sample-size reduction. Applying the observed inflation factor to the full-sample t-statistics reproduces the pattern of significance in the restricted sample (see Web Appendix C).

outcome for customers who received only targeted coupons. Because customers are randomly assigned to groups, $\hat{\beta}_F$ captures the difference $\bar{Y}(\text{Focal}) - \bar{Y}(\text{NoRandom})$, while $\hat{\beta}_{NF}$ captures the difference $\bar{Y}(\text{NonFocal}) - \bar{Y}(\text{NoRandom})$.

To improve estimation precision, we include (customer×date)-specific covariates as controls, which absorb residual variance in the purchase outcomes Y_i. The controls include RFM (Recency, Frequency, Monetary) measures calculated over a 90-day window prior to the coupon receipt. We also cluster standard errors at the customer level, because the same customer may appear in the sample on multiple dates (potentially leading to within-customer correlation across observations).

The regression framework supports the three outcome measures introduced in Table 2. In Section 6, we also use this approach to calculate category-level treatment effects. In particular, we estimate separate ATEs with category-level outcome measures for each campaign, and then average these ATEs across campaigns within each category. An alternative approach that pools observations across campaigns within a category and estimates a single category-level treatment effect, yields very similar estimates.

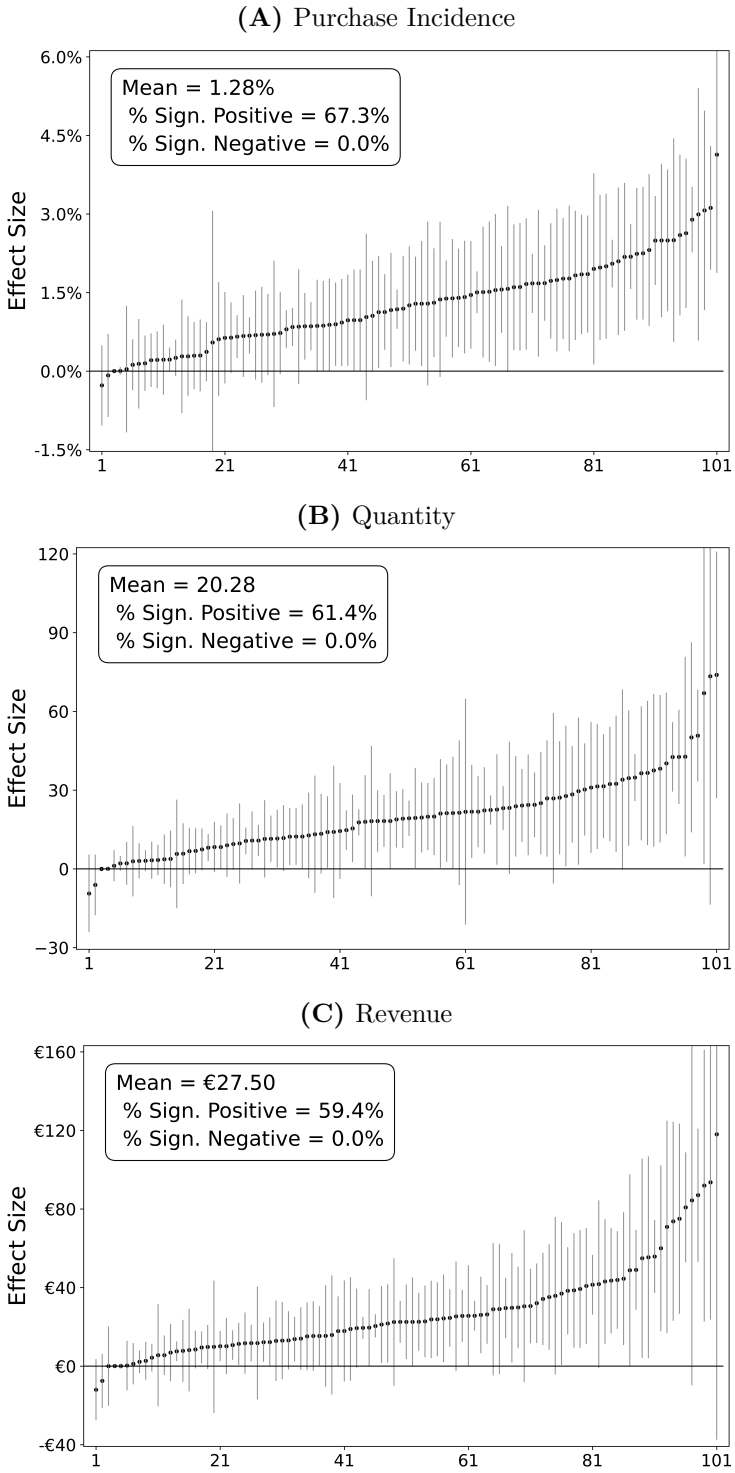
5. Brand Treatment Effects

This section evaluates the impact that store-entry coupons have on sales of the promoted (focal) brands. We first report the distribution of ATEs across the 101 CPG campaigns. We then investigate how these effects vary depending on customers' previous experience with the brand and category. The section concludes by analyzing how the treatment effects vary with the depth of the discount, offering insight into the primary mechanisms driving store-entry coupon effectiveness.

5.1 Distribution of Brand ATEs

In Figure 3, we report the distribution of the ATEs for the promoted brands. Each data point represents a treatment effect estimate for a single campaign on the day that the customer received the coupon, and the error bars indicate 95% confidence intervals. Panel A measures the percentage increase in purchase incidence, while Panels B and C report the incremental quantity (units) and revenue lift (Euros) per 1,000 coupons (per mille). For example, an effect size of €10 indicates that the focal brand's average revenue per 1,000 coupon impressions was €10 higher than the average revenue per 1,000 kiosk visits without a coupon for the focal brand.

Figure 3: Distribution of Average Treatment Effects (ATEs)



Notes: The figure summarizes the distribution of average treatment effects for the 101 brands. Panel A reports the incremental purchase incidence for each brand (in percentage points). Panel B reports the incremental quantity sold (in units per 1,000 coupons). Panel C reports the incremental revenue (in €per 1,000 coupons). Error bars indicate 95% confidence intervals.

Store-entry coupons drive incremental sales for almost all brands. The purchase incidence treatment effects are positive for 99 of the 101 campaigns, and 67.3% of the effects are statistically significant ($p < 0.05$). We observe a very similar pattern for the quantity and revenue treatment effects. Overall, the average ATE across the 101 brands is 1.28% incremental purchase incidence, 20.28 incremental units, and €27.50 additional revenue per 1,000 coupons. We do not find any negative treatment effects that are statistically significant for any campaign across any measure.

The revenue measure incorporates coupon discounts, which average 23.95% across the sample. To offset this price reduction, coupons need to increase purchase quantities by a factor of 1.31x to ensure that revenue breaks even. This is a substantial threshold, which might make some manufacturers hesitant to allocate budget to this channel. However, despite this breakeven threshold, store-entry coupons yield revenue gains for essentially every brand in our sample. Store-entry coupons drive a 4.9x average increase in purchase incidence, a 6.3x increase in quantity sold, and a 5.2x increase in revenues.

Table 3 illustrates these effects using six globally recognized brands. Store-entry coupons are effective in driving sales for all six brands. For example, Coke's revenue increases by 86%, and coupons yield even more substantial gains for other brands, with sales lifts for Maggi spice mixes exceeding 16x for quantity and 10x for revenue. As expected, revenue lifts are generally smaller than quantity lifts because the revenue measure accounts for the price reduction from the discount.

The Coke example is particularly interesting. Customers purchase Coke on 1.7% of shopping trips without Coke coupons, which is among the highest purchase rates for brands in our sample. This creates a risk that store-entry coupons provide discounts to customers who would purchase anyway. We find that despite the high baseline purchase rate, the incremental demand for Coke is large enough to offset the discounting.

To systematically investigate the variation in ATEs across the 101 campaigns, we analyze how treatment effects correlate with four brand features: penetration, loyalty, price position, and interpurchase time. These features are constructed using the retailer's loyalty card data. The strongest relationship is for brand penetration; brands with higher baseline penetration tend to yield larger ATEs. One explanation is that some customers do not need some categories, and store-entry coupons cannot change this. If few customers have pets, then we would expect both a low

Table 3: Treatment Effects for Six Global Brands

	Incidence		Quantity		Revenue	
	No Coupon	ATE	No Coupon	ATE	No Coupon	ATE
Ben's Original rice	0.14%	1.38%	1.74	18.25	€3.05	€26.34
Coke soft drinks	1.70%	1.29%	27.09	26.87	€26.00	€22.45
Johnnie Walker whisky	0.13%	0.97%	1.35	9.48	€12.81	€84.36
Lindt chocolates	0.11%	0.86%	1.47	10.83	€4.29	€24.57
Maggi spice mix	0.14%	1.40%	1.90	31.44	€2.19	€23.85
Schauma shampoo	0.08%	0.69%	0.84	9.00	€1.85	€13.03

Notes: The table reports the ATEs and outcomes in the condition without a random coupon for six globally recognized brands.

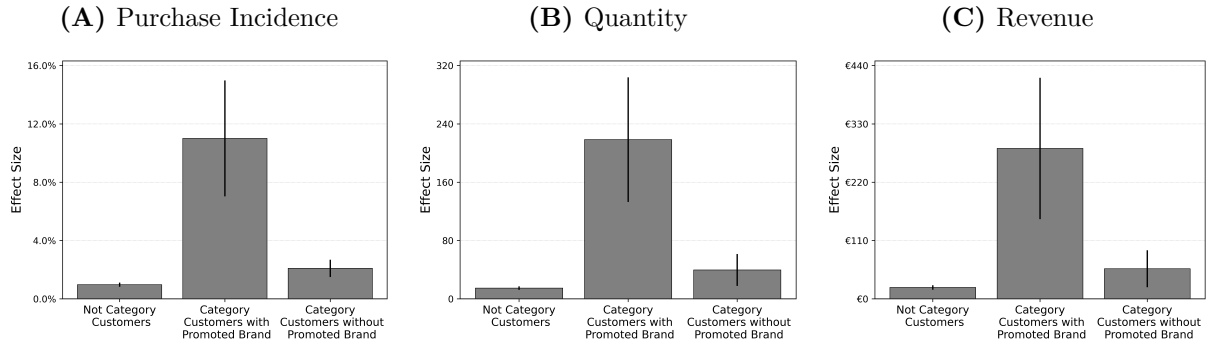
penetration rate for pet food brands and a relatively small response to pet food coupons. Other brand features are less predictive of the ATEs. We discuss these results in Web Appendix D, where we also provide precise definitions of the four brand features.

5.2 Heterogeneity in Treatment Effects Across Customers

Many retailers allow brands to choose whether store-entry coupons are distributed to past brand customers, customers of competing brands, or customers who have not recently purchased in the category. To investigate the variation in treatment effects across these options, we use loyalty data in the three months before the coupon experiment to group customers into three customer segments: *Not Category Customers* (80%), *Category Customers with Promoted Brand* (3%), and *Category Customers without Promoted Brand* (17%). We then calculate conditional average treatment effects (CATEs) separately for each segment, which we then average across the 101 brands. In Figure 4, we report the CATEs for the promoted brand (on the current trip) by each segment.

Figure 4 reveals an important distinction between segment-level responsiveness and the aggregate composition of the treatment effects. The largest treatment effects occur for the *Category Customers with Promoted Brand* segment. Among these customers, the promoted brand's revenue increased by over €283 per mille. However, significant lifts also appear among the customers who had only purchased other brands in the category in the previous three months (€57 per mille) or had not purchased in the category in that prior period (€21 per mille). Notably, while existing brand buyers are the most responsive per-coupon, they represent only 3% of the population.

Figure 4: Heterogeneity in Treatment Effects for the Promoted Brand



Notes: The figures report the average ATEs (on the current trip) averaged across the 101 focal campaigns. Each column represents a different customer segment. Error bars indicate 95% confidence intervals.

Because the *Category Customers without Promoted Brand* and *Not Category Customers* segments are substantially larger, they account for the vast majority of the aggregate revenue lift. These findings suggest that the value of store-entry coupons lies in their ability to also drive acquisition and category expansion, rather than just deepening loyalty among existing brand buyers.

5.3 Discount Depth

Recall that our experimental design randomized two coupon features. First, we varied which brand was assigned to individual customers so that one customer received a coupon for Coke, whereas another customer received a coupon for detergent. Second, we also randomized the discount level. For example, one customer received a 15% discount for Coke, whereas another customer received a 30% discount for Coke. We randomly assigned a discount with equal probability from the discount levels that the manufacturer made available for that brand. The exogenous variation in discount levels enables us to measure (causal) treatment effects as a function of discount depth.

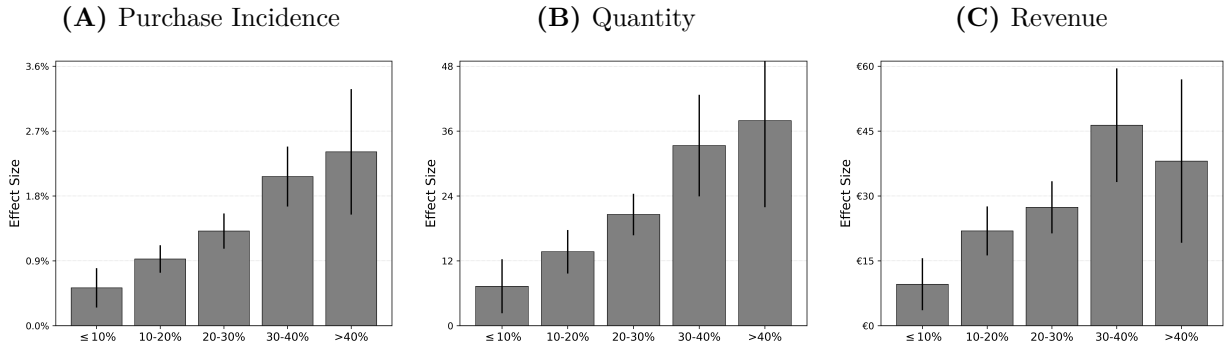
We start by segmenting observations by discount levels. We bin observations into five buckets each spanning 10 percentage points: $\leq$10%, (10%, 20%], (20%, 30%], (30%, 40%], and >40%. Using the identification strategy in Section 4, we calculate campaign-level treatment effects within each bucket and report the average ATEs in Figure 5.[7]

Figure 5 reports positive purchase incidence and quantity treatment effects for all discount levels (Panels A and B). Treatment effects increase monotonically with the discount depth, so that

[7]Some brands do not distribute coupons with the highest and lowest discount values. We present the results for the subset of brands that use all possible discount buckets in Web Appendix E.1.

larger discounts lead to larger treatment effects, but this trend flattens at the highest discount levels. The differences in treatment effects are statistically significant between small and large discounts, but not between the two largest discount bins. Previous research has reported similar patterns for coupon redemption rates (Dhar and Hoch 1996; Danaher et al. 2015; Molitor et al. 2016).

Figure 5: Treatment Effects at Each Discount Level



Notes: The figure depicts treatment effects as a function of the discount depth. Treatment effects within each discount bucket are first calculated at the brand level. The bars represent means calculated by averaging across the brands. Error bars indicate 95% confidence intervals calculated using a non-parametric bootstrap.

The revenue treatment effects are also all positive and statistically significant (Panel C). They increase monotonically for discounts up to 40%. However, for discounts above 40%, revenue treatment effects eventually start to decrease because the gains in purchase rates are offset by customers paying lower prices because of discounts (Neslin 1990).

The evidence that customers are more likely to purchase when they receive larger discounts is consistent with standard arguments that customers are price sensitive. They confirm that the discount itself is an important driver of the increase in sales of the focal brand. However, treatment effects in Figure 5 remain positive even at minimal discount levels (<10%). This suggests that the coupon also serves an "exposure" (or "advertising") function that increases brand salience independent of the price incentive. This is consistent with previous research that has argued that price promotions can enhance customer engagement and help to shape consumer behavior through psychological and social mechanisms (Shimp and Kavas 1984; Inman et al. 1990).

Although both mechanisms could contribute to coupon effectiveness, the price effect appears to be the primary driver of the total lift. If we interpret the treatment effect at the lowest discount

bin as a proxy for the exposure effect, the substantial and monotonic increase in lift as discounts move toward 40% indicates that the price component accounts for the majority of the incremental sales. We conclude that mere exposure to store-entry coupons may serve as a valuable point-of-sale "nudge", but significant shifts in brand performance remain largely dependent on the depth of the price incentive.

Because discount depth is randomly assigned, we can also evaluate its impact by simply comparing outcomes across discount levels within the same campaign. This comparison only uses the *Focal* observations: every customer in these comparisons received a random coupon, and the comparisons do not require any adjustments for targeted coupons. We report this analysis in Web Appendix E.2; the pattern of results is unchanged.

5.4 Discussion of Brand ATEs

There is a large increase in purchases of the promoted brand on the day customers received the coupon. The volume increase is large enough to offset the coupon discount, leading to revenue gains for nearly all 101 brands. The lift in revenue occurs even though store-entry coupons can be exercised immediately, which results in high redemption rates. In our empirical context, coupons are automatically redeemed when customers buy the promoted product. The revenue increase occurs despite this 100% "redemption rate".

Our findings highlight the versatility of store-entry coupons. Different brands have different objectives for their promotions. Some are focused solely on expanding top-line sales, while others want to induce brand switching or attract new customers into the category. Store-entry coupons are an effective tool for achieving these different goals. The sales increases for the promoted brand are not limited to the promoted brand's past customers. Sales of the promoted brand also increase for past customers of other brands, and for customers who had not recently purchased in the category. This suggests that store-entry coupons are as much an acquisition and expansion tool as they are a loyalty incentive.

The purchase incidence treatment effects increase monotonically with the depth of the coupon discounts. We interpret this as evidence that customer price sensitivity is a primary driver of these treatment effects. However, positive treatment effects at even nominal discount levels suggest a dual mechanism. While price incentives affect sales most significantly, the coupon's role as a point-

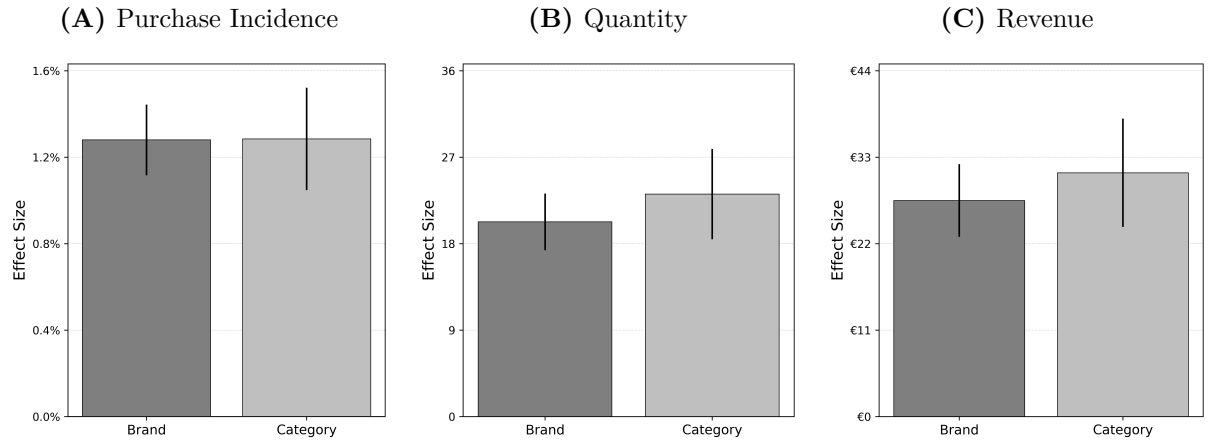
of-sale "nudge" likely enhances customer engagement and brand salience, providing a baseline lift independent of the discount magnitude.

6. Impact of the Store-Entry Coupons on Category Sales

While increasing a brand's sales is valuable for a manufacturer, retailers care about the performance of the entire category. This section shifts the focus from brand-level outcomes to evaluating whether store-entry coupons contribute to category expansion. We use the identification approach for category-level outcomes that we described at the end of Section 4.

Figure 6 provides consistent evidence of category expansion that matches the brand uplifts across all metrics. The increases in the *Quantity* outcome measures are particularly notable. The *Quantity* gains cannot be explained by price differences in the category, either due to the discounts in the treatment condition, or due to switching between higher or lower priced products in the category. The positive Revenue ATEs further confirm that the quantity increase more than compensates for the price reductions offered by the coupons. We provide a complete list of product categories and category-level ATEs in Web Appendix A.

Figure 6: Brand and Category ATEs

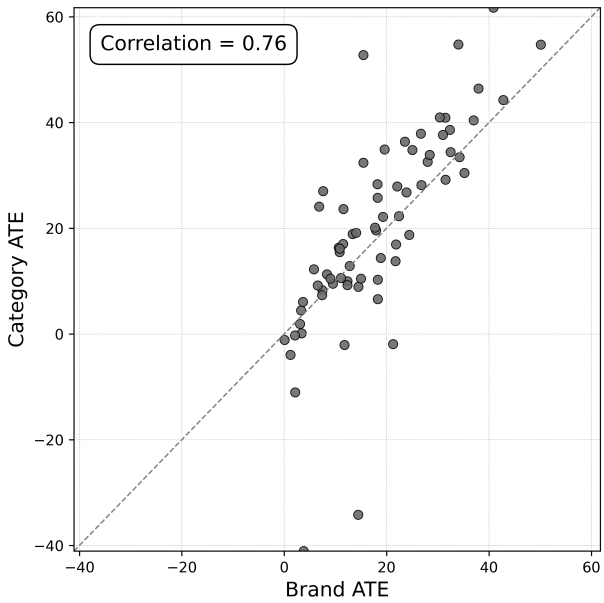


Notes: The figures report the average Brand and Category ATEs (on the current trip) averaged across the 71 product categories. Error bars indicate 95% confidence intervals.

Figure 7 illustrates the relationship between brand-level and category-level quantity ATEs. The strong pairwise correlation ($\rho = 0.76$, $p < 0.01$) suggests that campaigns driving the largest brand gains are also the most effective at expanding the category. The close alignment of the observations around the 45-degree line also highlights that the magnitude of the brand effects is

similar to the magnitude of the category effects. This implies that category expansion is a primary source of the incremental brand sales.

Figure 7: Scatter Plot of Brand ATEs and Category ATEs



Notes: The figure reports a scatter plot of brand-level ATEs (averaged by category) and category-level ATEs. All ATEs are measured using the *Quantity* outcome measure. The unit of observation is a category.

In the remainder of this section, we investigate the drivers and boundary conditions of this category expansion. To rule out price effects, we focus our analysis on *Quantity* as the outcome measure.

6.1 Temporal Substitution

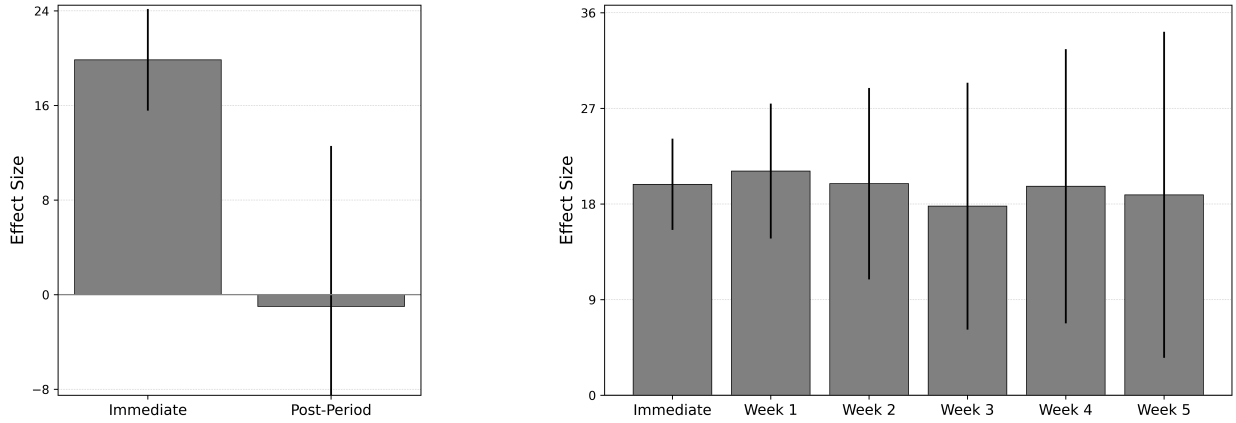
A common concern with price promotions is that they can result in a "post-promotion dip": immediate sales gains are offset by reduced demand in subsequent weeks (see, for example, Neslin et al. 1985; Neslin and Schneider Stone 1996; Van Heerde et al. 2000; Hendel and Nevo 2003; Macé and Neslin 2004). To investigate whether store-entry coupons pulled demand forward resulting in a post-promotion dip, we repeated our category-level analysis using longer measurement windows that combine outcomes on the current visit with outcomes on future visits.

In Figure 8, we report the post-promotion ATEs using two different approaches. Panel A summarizes outcomes for the "Immediate" ATE on the day of the coupon distribution, and the 5-week period starting the day after the distribution date ("Post-Period"). The "Immediate" ATE

on the day of coupon distribution is positive and significant, while the "Post-Period" ATE (the following 5 weeks) is essentially zero.

Panel B reports cumulative findings using six measurement windows: the day the coupon is distributed ("Immediate"), the distribution date and the next six days ("Week 1"), the distribution date and the next thirteen days ("Week 2"), etc. This cumulative analysis shows no evidence of erosion in the initial sales lift over time, suggesting that the observed category expansion primarily reflects incremental demand rather than simple purchase acceleration.[8]

Figure 8: Temporal Substitution: Immediate and Post-Period Category-Level ATEs



(A) Distinct Periods

(B) Cumulative Periods

Notes: These figures plot category-level *Quantity* ATEs averaged across the 71 categories. In Panel (A), we report separate findings for the Immediate and Post-Period. In Panel (B), we report cumulative findings. Error bars indicate 95% confidence intervals.

While the aggregate effect shows no dip, we tested whether specific categories prone to stockpiling behaved differently. Using the Narasimhan et al. (1996) two-item *Stockpiling* scale, we found no significant relationship between a category's suitability for storage and its Immediate or Post-Period ATE. We do observe an indication of a post-promotion dip in suburban stores, where stockpiling is presumably more common. In urban stores, it is clear that coupon effectiveness extends beyond any stockpiling effects. These results are reported in Appendix C, and the data collection for the stockpiling variable is described in Web Appendix F.1.

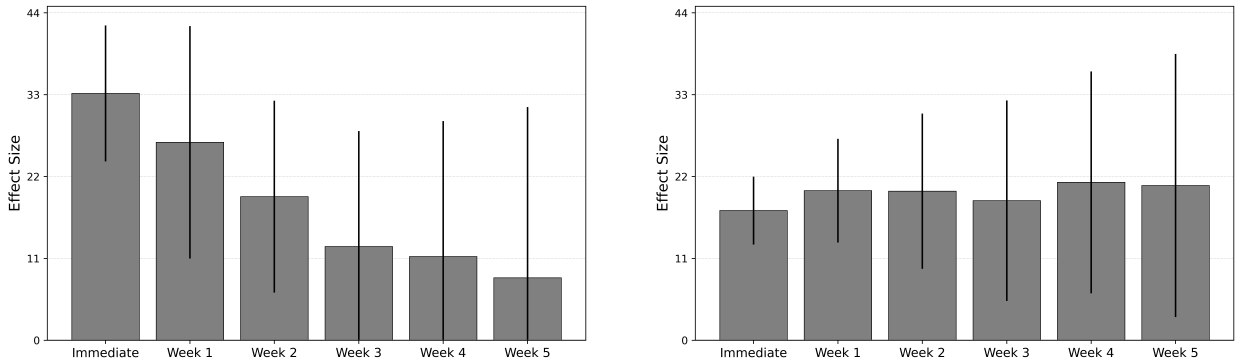
To explore this further, we compared perishable versus non-perishable categories. The 11

[8] At the category level, post-promotion dips remain significant only for two of the 71 categories (using Bonferroni correction): alcohol chocolates and potato pancakes.

perishable categories include ready-to-eat meals, dairy products, and (cooked) meat sausages (see a complete list in Web Appendix A). Intuitively, shelf-stable goods should be more susceptible to stockpiling and thus more likely to show a post-promotion dip. However, Figure 9 reveals the opposite: in perishable categories (e.g., milk, yogurt, sausages), the immediate gains are almost completely offset in the Post-Period. In contrast, non-perishable categories show no such erosion, with the overall effect remaining stable over time.

This divergence suggests that the post-promotion dip in our setting may be driven by consumption shifts rather than household inventory behavior. An immediate surge in consumption (e.g., eating more chocolate) may lead to subsequent satiation or a desire for variety, reducing demand in the following weeks. For shelf-stable items, customers may maintain large enough inventories to buffer these short-term consumption fluctuations. However, in perishable categories, purchase timing is tightly linked to consumption. While we do not directly observe consumption, products that can be consumed immediately often belong to high-impulse categories (e.g., chocolate). In Web Appendix F.2, we observe a negative correlation between Post-Period ATEs and measures of whether the category is susceptible to impulse purchasing (including the two-item impulse survey scale by Narasimhan et al. 1996). This correlation is not statistically significant.

Figure 9: Category ATEs: Perishable vs. Non-Perishable Categories



(A) Perishable Categories **(B)** Non-Perishable Categories

Notes: The figures report cumulative category-level *Quantity* ATEs averaged across the 11 perishable categories (Panel A) and 60 non-perishable categories (Panel B). Error bars indicate 95% confidence intervals.

One caveat is that a null Post-Period effect could reflect offsetting forces. In particular, a coupon-induced purchase could trigger additional targeted coupons for the focal brand on later visits, which could mask a post-promotion dip. The data do not support this explanation: across

campaigns, the subsequent coupon delivery shows no difference between customers who did and customers who did not receive a random coupon for the focal brand in the experiment (Web Appendix G). Post-Period effects are therefore unlikely to reflect the retailer's targeting algorithm.

In summary, while we do find some examples of temporal substitution, this mechanism cannot fully explain the category expansion we observe on the day the store-entry coupons are received. When averaging across all 71 categories, the store-entry coupons did not result in a post-promotion dip. We next investigate a different explanation for category expansion: customers switching purchases from competing retailers.

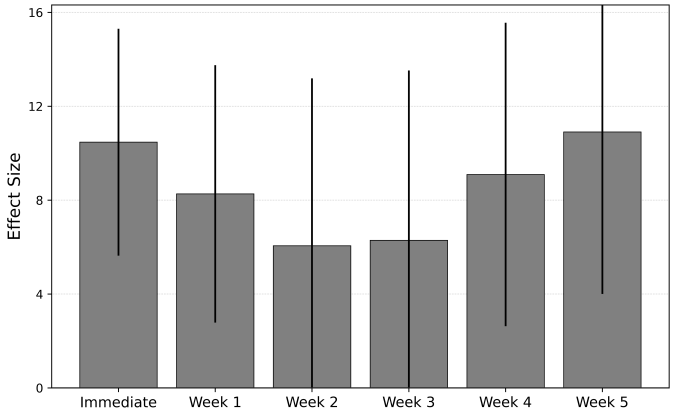
6.2 Store Switching

A possible explanation for category expansion in our setting is store switching. Customers routinely split grocery purchases across several retailers, so a store-entry coupon received at the start of a shopping trip can shift into the current basket purchases they would otherwise have made at a competing retailer (on a future trip). The customer purchases at the focal retailer now with a certain discount, rather than purchasing at another retailer in the future, where the availability of a discount is uncertain.

In Figure 10, we investigate a setting in which store switching is the most likely explanation for incremental purchases at the focal retailer. Specifically, we select 13 categories in which consumption expansion is very unlikely: deodorants, facial tissues, hair sprays, paper towels, shampoo, shower gels, toilet paper, toothpaste, dishwasher detergents, fabric softeners, laundry detergents, cat food, and dog food. A coupon for toothpaste or pet food is unlikely to induce a customer to use more of the product or acquire a new pet, and so category expansion cannot result from consumption expansion. We further focus on customers who did not purchase the category at the focal retailer in the three months before the experiment. If these customers purchase the fixed-consumption categories, it appears that they do so at other retailers. Incremental purchases for these categories and by these customers at the focal retailer would indicate store switching. Figure 10 is consistent with this prediction: immediate category expansion at the focal retailer, with a significant effect even after five weeks.

We provide two additional analyses to support the store switching explanation. First, we compare treatment effects calculated using purchases at stores located close to competing stores

Figure 10: Fixed-Consumption Categories and Customers Without Category Purchase



Notes: The figure reports cumulative category-level *Quantity* ATEs averaged across 13 categories in which we do not expect coupons to cause changes in consumption. The analysis focuses on customers without a prior category purchase at the focal retailer in the 90 days before customers received the store-entry coupon. Error bars indicate 95% confidence intervals.

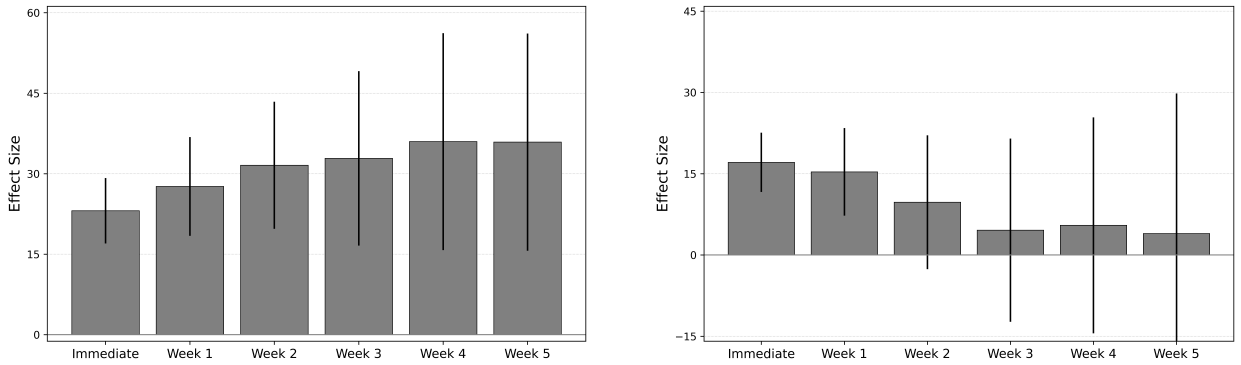
versus stores farther from competitors. If store switching plays an important role, its impact should vary with the geographic density of the retail stores. Specifically, in focal stores located near competitors, we expect (a) larger immediate category expansion and (b) a smaller post-promotion dip at the focal retailer, as the "dip" is effectively externalized to the competitor's store. To test this, we divided our stores into two groups based on a median split of the distance to the nearest competing supermarket.[9]

The findings in Figure 11 support these predictions. In stores located near competitors (Panel A), we observe robust category expansion that persists over the extended measurement window. This pattern is consistent with customers shifting purchases from nearby competing stores in response to the store-entry coupon. Conversely, in stores farther from competitors (Panel B), the initial category expansion is followed by a notable erosion of the effect. In these more isolated stores, the initial surge in demand is more likely to reflect substitution from the focal retailer's own future sales. In Web Appendix J, we further decompose these patterns. When we restrict the analysis to suburban locations, the differences in immediate and post-promotion outcomes are significantly different for focal stores located close to and far from competitors. In urban areas,

[9]Distance to the nearest competing supermarket is calculated using geographic coordinates (latitude and longitude). We obtain the coordinates for all focal and competing stores, compute the Haversine distance between each focal store and every competing store, and define the distance to the nearest competitor as the minimum of these distances.

the differences are not statistically significant, perhaps because the distance to competitor stores is hard to interpret with customers using different transportation modes. These findings are consistent with Taylor and Hollenbeck (2021), who report larger store switching effects in areas with lower population density.

Figure 11: Competing Stores Close or Far From Focal Retailer's Stores



(A) Competing Stores Close **(B)** Competing Stores Far

Notes: The figures report category-level *Quantity* ATEs averaged across the 71 categories. In Panel (A), we calculate ATEs using purchases at the focal retailer's stores located close to competing stores. In Panel (B), we calculate ATEs using purchases at stores located far from competing stores. Error bars indicate 95% confidence intervals.

We further investigate store switching using household panel data. We obtained panel data covering purchases from grocery retailers for a sample of German households. We first use the panel data to confirm that customers shop at multiple retailers. The median household shopped at nine distinct retail chains, and 98% purchased at three or more competing chains. Even among the 25% of customers with the highest share-of-wallet at the focal retailer, 95% shopped at three or more competing chains (Web Appendix I).

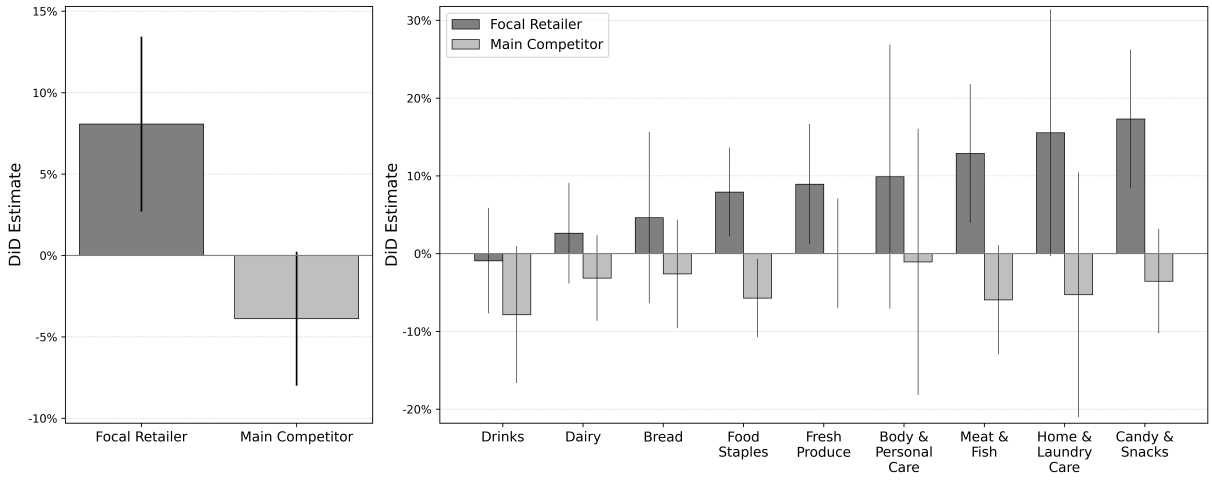
The focal retailer initially introduced the coupon system in only one geographic region, and the panel data captures shopping behavior in the retailer's stores with and without the coupon system. Moreover, the time period covered by the panel data span the introduction of the system. This provides a quasi-experimental setting to compare sales at the focal retailer before and after the system's introduction, using the stores without the system as a control group.

We conduct a difference-in-differences (DiD) analysis separately for each category to estimate the impact that the coupon system had on sales. We also repeat the analysis for the retailer's main

competitor to determine whether its sales moved in the opposite direction. Additional details about this DiD analysis are reported in Web Appendix H, where we also include a replication using a placebo date.

Figure 12 reports the impact of the new coupon system on sales. The left panel shows overall sales, while the right panel breaks down the results across the nine product groups used as category definitions (as provided by the panel data supplier). The DiD analysis shows that the introduction of the coupon system led to an increase in the focal retailer's sales ($p < 0.01$), while simultaneously decreasing sales at the retailer's main competitor ($p = 0.08$). These findings are consistent with store switching; the coupon system caused customers to switch purchases from the (main) competing retailer to the focal retailer.

Figure 12: Diff-in-Diff Coefficients From Panel Data

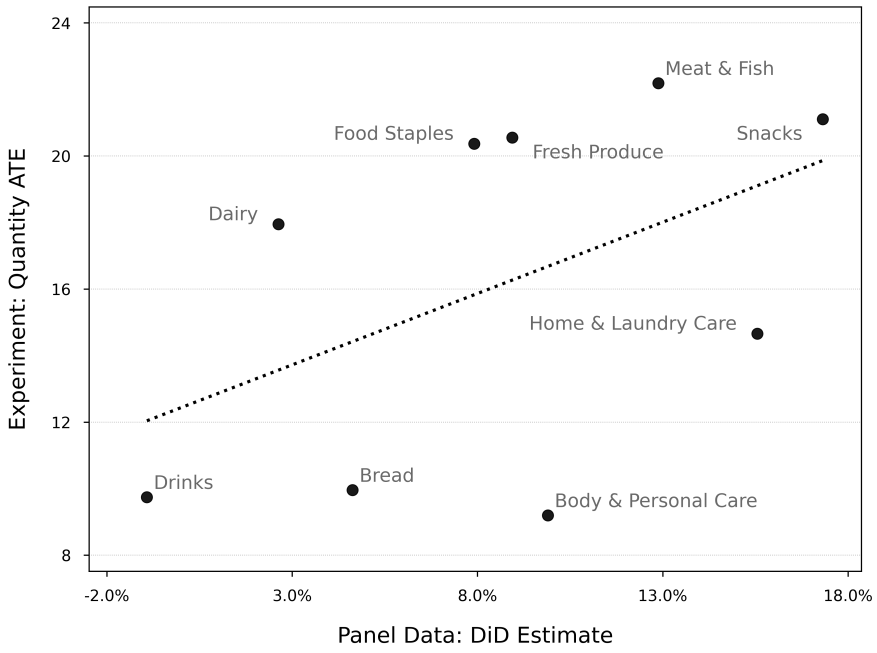


Notes: The figures report diff-in-diff coefficients from the panel data. Error bars indicate 95% confidence intervals. See Web Appendix H for additional details regarding the panel data DiD analysis.

In additional analysis, we investigate whether the categories exhibiting the largest growth in sales (at the focal retailer) in the panel data correspond to those with the largest ATEs in our experiment. In particular, we matched the categories involved in our experiments with the nine product groups in the panel data and averaged the (immediate) category expansion ATEs across the categories within each group. In Figure 13, we compare the panel data DiD coefficients within each product group with the average category expansion ATEs from our experiments.

We observe a positive relationship across the product categories. Product groups for which

Figure 13: ATEs from Experiments and Diff-in-Diff Coefficients from the Panel Data



Notes: The figure reports a scatter plot of category-level ATEs from the experiments and diff-in-diff coefficients from the panel data. The nine product groups are defined by the supplier of the panel data.

the store-entry coupon system caused switching from competing retailers to the focal retailer (e.g., candy and snacks) are the same categories in which we observe the largest category expansion effects in our experiment. Across the nine groups, the (rank-order) correlation between the panel data DiD coefficients and the average experimental ATEs is 0.47.

What makes this relationship especially notable is that the results are obtained from completely separate datasets and with different methodologies. Our field experiment captures immediate purchases by customers who are already in the focal store, while the panel data reflects long-term behavior across multiple store visits. In our experiments, we measure the impact of randomly assigned coupons in the focal category, while the panel data measures the impact of (primarily) targeted coupons. Despite these differences, both studies provide convergent evidence of store switching.

6.3 Discussion of Category Expansion

Our category-level analysis provides consistent evidence that store-entry coupons expand category sales at the focal retailer on the day customers receive them. The effect appears across purchase incidence, quantity, and revenue, and its magnitude closely tracks the lift in the promoted brand,

so the store-entry coupons grow the category rather than merely switch brand shares within it.

We investigated the source of the increase in category sales, and found no evidence for several standard explanations. Brand switching is largely ruled out by construction, since we focus on category-level Quantity, which nets out substitution among brands. We find no evidence of forward-buying or stockpiling in the aggregate analysis. The Post-Period effect is essentially zero, the cumulative lift does not erode over the following weeks, and categories that are more storable do not exhibit a larger post-promotion dip. Where a post-promotion dip does appear, it is concentrated in perishable categories, the opposite of what household stockpiling would predict, pointing to short-run shifts in consumption timing rather than stockpiling. Temporal substitution appears in isolated cases but cannot explain the category expansion we observe.

The evidence instead points to store switching. Grocery shoppers multi-home extensively, with the median panel household buying from nine chains. A store-entry coupon can pull a purchase into the current basket that the customer would otherwise have made at a competitor later. Three findings support this interpretation. First, we find category expansion in categories with inelastic consumption among customers who had not purchased these categories at the focal retailer in the prior three months. Second, we compare focal stores located near and far from competitors. There is immediate category expansion in both groups of stores, but over the following five weeks, the effect survives in stores located near competitors but erodes in stores that are far from competing stores. Finally, a difference-in-differences analysis of external panel data shows that introducing the coupon system raised the focal retailer's category purchases, while lowering the main competitor's, with the largest panel effects in the same categories that show the largest expansion in the experiment. While the data does not provide direct evidence of store switching, our empirical findings are complementary, and the store switching interpretation is the most consistent with the results.

7. Conclusions

A long-standing challenge for retailers is to find an instrument for delivering promotions to customers when they are about to purchase (at the bottom of the purchase funnel). While digital channels have solved this problem through sponsored search and on-site advertising, physical retail has historically resorted to direct mail, newspaper FSIs, and checkout coupons, which are received

either before or after a shopping trip. By distributing coupons during shopping trips, in-store kiosks and mobile apps represent a fundamental shift in this landscape.

We investigate the effectiveness of in-store promotions using data from a large-scale field experiment that randomly varied coupon distribution within a business-as-usual environment. Coupons were distributed via kiosks, which loyalty program members typically visited at the start of their trips. The experimental data provide a causal measure of the impact of store-entry coupons on customer purchasing. We summarize the key findings in Table 4.

Table 4: Summary of Key Findings

Promoted Brand Sales: Store-entry coupons substantially increase sales for promoted brands at the focal retailer. The incremental unit sales offset the price discounts, so that revenue increases for nearly all campaigns.
Practical Versatility: Store-entry coupons can be used to increase top-line sales, induce trial among competitors' customers, and attract new customers to the category.
Category Expansion: Store-entry coupons lead to category expansion, with the magnitude of the category expansion closely matching the increase in sales of the focal brand.
Temporal Substitution: Temporal substitution occurs in some categories but does not appear to stem from stockpiling and is not sufficient to explain the broader category expansion observed across the study.
Store Switching: Store switching appears to contribute to category expansion, with customers purchasing immediately at the focal retailer instead of purchasing at competing retailers (on future visits).

Store-entry coupons are remarkably effective, causing a nearly fivefold increase in sales for promoted brands on average. The sales lifts are large enough to offset the price discounts, resulting in a net revenue increase for nearly all brands. The promotions successfully reactivated past brand customers, induced trial among competitors' customers, and attracted new customers to the category. This makes store-entry coupons a versatile marketing tool that can be used to both retain past customers and attract new customers.

The impact of store-entry coupons extends beyond the promoted brand to drive significant category expansion at the focal retailer. Across the brands, the size of the focal brand effect closely aligns with the magnitude of the category expansion, suggesting a direct relationship between the two. Our analysis investigates mechanisms that contribute to category expansion. In some categories, we find evidence of temporal substitution. However, this is not driven by traditional

stockpiling, but rather represents a *shift in consumption*, where customers consume more now and less in the future. The explanation for category expansion that is most consistent with our empirical findings is *store switching*. Customers purchase immediately from the focal store instead of making purchases at competing retailers on future visits.

Our findings have different implications for manufacturers and retailers. In our setting, as is standard in retail media, the discounts are funded by the promoted brands rather than by the retailer. For the retailer, the economically relevant question is therefore not the discount depth versus its own margin, but whether the coupons expand the category or merely reshuffle brand shares. Our results show that the sales lift is predominantly category expansion, and this incremental volume arrives at full margin. To the extent that store switching drives the category expansion, the results are *positive* for retailers.

For the brands funding the discounts, the relevant margin is the manufacturer's. Gross margins of large CPG manufacturers average approximately 47% of manufacturer revenue, which is equivalent to approximately 25% off the retail price. [10] A 23.95% average discount off the retail price is therefore large relative to the gross margin earned on any single unit, and the profitability of funding these coupons depends on how many of the discounted purchases are incremental. Our treatment effects measure exactly this incrementality. However, the store-switching evidence adds a caution: part of a brand's lift at the focal retailer is volume relocated from competing retailers, which is incremental for the retailer but not necessarily for the brand market-wide. A brand may be paying a discount to move its own sales between stores. We also note that treatment effects are significant even with discounts below 10%, so deep discounting is a design choice rather than a requirement of the instrument.

Limitations and Future Research

A limitation in our study is that we do not observe the wholesale prices that the retailer pays to manufacturers for their brands. This means we are unable to report how store-entry coupons affect the incremental profits earned by the retailer or the manufacturers. Retailers and manufacturers

[10] We approximate these margins using publicly reported financial statements from 2018–2022. Ten large supermarket retailers report cost of goods sold averaging approximately 75% of retail revenue, and ten of the largest CPG manufacturers report cost of goods sold averaging approximately 53% of manufacturer revenue. Ahold Delhaize separately discloses that payments to vendors represent approximately 72% of its cost of goods sold. Together, these figures imply that the wholesale price averages approximately 54% of the retail price, so a gross margin of 47% of manufacturer revenue is equivalent to approximately 25% of the retail price.

could combine the treatment effects reported in this paper with their own wholesale price data to estimate changes in profits. A related question is how store-entry coupons increase overall channel profits, and how the incremental profits are shared between retailers and manufacturers. Quantifying how marketing actions affect total brand market shares across retailers is also an important direction for future research.

A feature of store-entry coupons is that their effectiveness is limited to the customers who actually receive them; in our setting, loyalty program members who visit the kiosk at the start of a shopping trip. We offer two perspectives on this limitation. First, the (very) positive outcomes for the retailer in this study suggest we will continue to see innovations in distribution technologies, so that over time, a larger proportion of customers will receive store-entry coupons. Second, the incomplete adoption of the kiosk system may also be a valuable feature of the system. Like other retailers, this retailer intentionally uses its loyalty program to price discriminate. The coupons allow the retailer to charge lower effective prices to customers who have signed up for the loyalty program. Price discrimination is only effective when some customers do not get the discounts.

Our data describe the response to CPG coupons delivered via in-store kiosks. Many retailers have developed alternative in-store media channels to deliver promotions to customers, including retail apps, in-store video, and displays. Examples can be found in consumer electronics (Best Buy), department stores (Saks), fast food (Starbucks), home improvement (Home Depot), beauty (Ulta), and other retailers. Future research could study the performance of retail media in these settings.

The panel dataset we use to study store switching is large enough to investigate changes in aggregate purchases by product group, but it is not large enough to detect store switching at the brand level. Data limitations are a common obstacle to measuring store switching. The lack of accurate market-level data means that store switching is often simply ignored in the promotions literature. Additional research investigating how promotions impact customer store visits and their purchases at different retailers would be a valuable contribution to this literature.

References

Abratt R, Goodey SD (1990) Unplanned buying and in-store stimuli in supermarkets. *Managerial and Decision Economics* 11(2):111–121.

Ailawadi KL, Harlam BA, Cesar J, Trounce D (2006) Promotion profitability for a retailer: the role of promotion, brand, category, and store characteristics. *Journal of Marketing Research* 43(4):518–535.

Ailawadi KL, Neslin SA (1998) The effect of promotion on consumption: Buying more and consuming it faster. *Journal of Marketing Research* 35(3):390–398.

Angrist JD, Imbens GW, Rubin DB (1996) Identification of causal effects using instrumental variables. *Journal of the American Statistical Association* 91(434):444–455.

Bawa K, Shoemaker RW (1987) The effects of a direct mail coupon on brand choice behavior. *Journal of Marketing Research* 24(4):370–376.

Bell DR, Chiang J, Padmanabhan V (1999) The decomposition of promotional response: An empirical generalization. *Marketing Science* 18(4):504–526.

Berthiaume D (2022) Survey: Online shoppers search for coupons. https://chainstoreage.com/survey-online-shoppers-search-coupons, accessed: 2025-01-16.

Bucklin RE, Lattin JM (1991) A two-state model of purchase incidence and brand choice. *Marketing Science* 10(1):24–39.

Chiang J (1991) A simultaneous approach to the whether, what and how much to buy questions. *Marketing Science* 10(4):297–315.

Chiang J (1995) Competing coupon promotions and category sales. *Marketing Science* 14(1):105–122.

Chiou-Wei SZ, Inman JJ (2008) Do shoppers like electronic coupons?: A panel data analysis. *Journal of Retailing* 84(3):297–307.

Danaher PJ, Smith MS, Ranasinghe K, Danaher TS (2015) Where, when, and how long: Factors that influence the redemption of mobile phone coupons. *Journal of Marketing Research* 52(5):710–725.

Dekimpe MG, Hanssens DM, Silva-Risso JM (1998) Long-run effects of price promotions in scanner markets. *Journal of econometrics* 89(1-2):269–291.

Dhar SK, Hoch SJ (1996) Price discrimination using in-store merchandising. *Journal of Marketing* 60(1):17–30.

Fok D, Horváth C, Paap R, Franses PH (2006) A hierarchical bayes error correction model to explain dynamic effects of price changes. *Journal of Marketing Research* 43(3):443–461.

Gabel S, Simester D, Timoshenko A (2024) How retailers became ad platforms. Harvard Business Review, URL https://hbr.org/2024/06/how-retailers-became-ad-platforms, accessed: 2025-01-16.

Grewal D, Ahlbom CP, Noble SM, Shankar V, Narang U, Nordfält J (2023) The impact of in-store inspirational (vs. deal-oriented) communication on spending: The importance of activating consumption goal completion. *Journal of Marketing Research* 60(6):1071–1094.

Grover R, Srinivasan V (1992) Evaluating the multiple effects of retail promotions on brand loyal and brand switching segments. *Journal of marketing research* 29(1):76–89.

Gupta S (1988) Impact of sales promotions on when, what, and how much to buy. *Journal of Marketing research* 25(4):342–355.

Hamstra M (2023) Stop & shop pilots kiosks for coupons and more. https://www.supermarketnews.com/finance/stop-shop-pilots-kiosks-for-coupons-and-more, accessed: 2025-01-16.

Heilman CM, Nakamoto K, Rao AG (2002) Pleasant surprises: Consumer response to unexpected in-store coupons. *Journal of Marketing Research* 39(2):242–252.

Hendel I, Nevo A (2003) The post-promotion dip puzzle: what do the data have to say? *Quantitative Marketing and Economics* 1:409–424.

Herhausen DB, Benoit S, Gabel S, de Jong D, Skiera B (2026) Retail media beyond the hype: A framework for value creation *(Working Paper)*.

Hoban PR, Bucklin RE (2015) Effects of internet display advertising in the purchase funnel: Model-based insights from a randomized field experiment. *Journal of Marketing Research* 52(3):375–393, URL http://dx.doi.org/10.1509/jmr.13.0277.

Hui SK, Inman JJ, Huang Y, Suher J (2013) The effect of in-store travel distance on unplanned spending: Applications to mobile promotion strategies. *Journal of Marketing* 77:1–16.

Inman JJ, McAlister L, Hoyer WD (1990) Promotion signal: proxy for a price cut? *Journal of consumer research* 17(1):74–81.

Johnson GA (2023) Inferno: A guide to field experiments in online display advertising. *Journal of Economics & Management Strategy* 32(3):469–490.

Johnson GA, Lewis RA, Nubbemeyer EI (2017) Ghost ads: Improving the economics of measuring online ad effectiveness. *Journal of Marketing Research* 54(6):867–884.

Lim J, Currim IS, Andrews RL (2005) Consumer heterogeneity in the longer-term effects of price promotions. *International Journal of Research in Marketing* 22(4):441–457.

Macé S, Neslin SA (2004) The determinants of pre-and postpromotion dips in sales of frequently purchased goods. *Journal of Marketing Research* 41(3):339–350.

Mela CF, Jedidi K, Bowman D (1998) The long-term impact of promotions on consumer stockpiling behavior. *Journal of Marketing Research* 35(2):250–262.

Molitor D, Reichhart P, Spann M (2016) Location-based advertising: Measuring the impact of context-specific factors on consumers' choice behavior. *In Proceddings of the Thirty Seventh International Conference on Information Systems (ICIS)* .

Narasimhan C, Neslin SA, Sen SK (1996) Promotional elasticities and category characteristics. *Journal of Marketing* 60(2):17–30.

Neslin SA (1990) A market response model for coupon promotions. *Marketing Science* 9(2):125–145.

Neslin SA, Henderson C, Quelch J (1985) Consumer promotions and the acceleration of product purchases. *Marketing Science* 4(2):147–165.

Neslin SA, Schneider Stone LG (1996) Consumer inventory sensitivity and the postpromotion dip. *Marketing Letters* 7:77–94.

Nijs VR, Dekimpe MG, Steenkamp JBE, Hanssens DM (2001) Tracing the impact of price promotions across categories. *Tilburg University the Netherlands Working paper* .

Osuna I, González J, Capizzani M (2016) Which categories and brands to promote with targeted coupons to reward and to develop customers in supermarkets. *Journal of Retailing* 92(2):236–251.

Pollay RW (1968) Customer impulse purchasing behavior: A reexamination. *Journal of Marketing Research* 5(3):323–325.

Raju JS, Dhar SK, Morrison DG (1994) The effect of package coupons on brand choice. *Marketing Science* 13(2):145–164.

Rook DW (1987) The buying impulse. *Journal of Consumer Research* 14(2):189–199.

Rubin DB (1974) Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology* 66(5):688.

Shapiro MF (2001) Impulse purchases. *NPN, National Petroleum News* 93(7):40–40.

Shimp TA, Kavas A (1984) The theory of reasoned action applied to coupon usage. *Journal of consumer research* 11(3):795–809.

Statista (2026) Companies with highest digital advertising revenues worldwide in 2025. URL www.statista.com/statistics/1285405/revenues-digital-ad-major-internet-companies, accessed: 2026-02-24.

Taylor W, Hollenbeck B (2021) Leveraging loyalty programs using competitor based targeting. *Quantitative Marketing and Economics* 19(3):417–455, ISSN 1573-711X, URL http://dx.doi.org/10.1007/s11129-021-09237-y.

Van Heerde HJ, Leeflang PS, Wittink DR (2000) The estimation of pre-and postpromotion dips with store-level scanner data. *Journal of Marketing Research* 37(3):383–395.

Venkatesan R, Farris PW (2012) Measuring and managing returns from retailer-customized coupon campaigns. *Journal of Marketing* 76(1):76–94.

Zhou L, Wong A (2004) Consumer impulse buying and in-store stimuli in chinese supermarkets. *Journal of International Consumer Marketing* 16(2):37–53.

Appendix A. Sample Selection Criteria

Our analysis uses two sample selection criteria. The first criterion focuses on estimation precision. We restrict our analyses to the brands that had regular sales prior to the experiment and distributed at least 400 random coupons. This criterion eliminates new products and short campaigns (e.g., those lasting only one or two days).

The second sample selection criterion recognizes that brands could define eligibility criteria for their coupons. For example, a pet food brand only distributed coupons to customers who had purchased pet food before. Random coupons were not distributed to non-eligible customers, so we cannot measure *average treatment effects* for these brands. Not all brands used eligibility criteria, and the retailer did not record the eligibility criteria defined by brand managers.

We detect which brands had eligibility criteria by comparing the pretreatment variables across experimental conditions. In particular, we regress coupon assignments on three pretreatment variables, and identify brands for which pretreatment variables are not jointly significant in an F-test with a 95% confidence level. The pretreatment variables include the number of store visits, revenue, and the number of unique products purchased during the 90 days prior to the experiment (from August to October 2015). For each brand, we estimate two regression models that compare assignment into *Focal vs. NonFocal* and *Focal vs. NoRandom* (see Section 4), and we require that neither F-test is statistically significant.

Appendix B. Identification of Treatment Effects

Recall that in the main text, we define an *observation* as a customer×date combination, and we restrict our analysis to customer×date combinations in which the customer used the kiosk system (at least once). We denote a customer×date combination by $i \in 1, .., N$. We also define $Y_i(1)$ and $Y_i(0)$ as potential outcomes for a customer×date observation i, with and without a coupon for the focal brand (Rubin 1974). Our goal is to estimate the average treatment effect (ATE): $\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$.

The proposed identification approach relies on two assumptions:

Assumption 1 (Stable Unit Treatment Value Assumption; *SUTVA*): The potential outcomes for an observation do not vary with the treatment assignments for other observations.

Assumption 2: Purchases of a focal brand are not affected by coupons from other brands.

Our first assumption recognizes that the same customer could appear in multiple customer×date observations in our data, and we treat these observations as independent. The second assumption implies that only focal coupons are relevant for estimating ATEs (instead of all seven coupons on the printout). Replacing a focal coupon with a random non-focal coupon moves from a potential outcome $Y_i(1)$ to a potential outcome $Y_i(0)$. These assumptions are standard in the causal inference literature.

Proposition: The following estimator provides an unbiased estimate of the average treatment effect for the focal brand:

$$\text{ATE} = \bar{Y}(\text{Focal}) - \left[\bar{Y}(\text{NoRandom}) - 1/p \cdot \left(\bar{Y}(\text{NoRandom}) - \bar{Y}(\text{NonFocal})\right)\right] \tag{B.1}$$

where $\bar{Y}(\cdot)$ indicates the average outcome in the corresponding group, and p captures the probability that a random *non-focal* coupon replaces a targeted *focal* coupon in the NonFocal_b group.

Proof. We can estimate the first term in the ATE as an average monetary outcome in the Focal condition. Focal observations are representative of the customer population, and in our experiment, they always received a coupon for the focal brand:

$$\mathbb{E}[Y(1)] = \mathbb{E}[Y_i \mid \text{Focal}] = \bar{Y}(\text{Focal}) \tag{B.2}$$

To estimate the expected potential outcome without a focal coupon, $\mathbb{E}[Y_i(0)]$, it is helpful to consider a group NoFocal, in which all targeted coupons for a focal brand are replaced by random coupons. Our data does not contain a group NoFocal. However, if we had such data, we could estimate $\mathbb{E}[Y_i(0)]$ as follows:

$$\mathbb{E}[Y_i(0)] = \mathbb{E}[Y_i \mid \text{NoFocal}] = \bar{Y}(\text{NoFocal})$$

The distribution of outcomes in NonFocal is a mixture of outcomes from NoFocal and outcomes from NoRandom, where a share of NoFocal observations is p. Indeed, in the NonFocal group, random coupons replace share p of targeted coupons in the printouts, and the remaining random coupons replace targeted coupons for non-focal brands, which does not change the outcomes for the focal brand. We can thus estimate $\mathbb{E}[Y_i(0)]$ as follows:

$$\mathbb{E}[Y_i \mid \text{NonFocal}] = p \cdot \mathbb{E}[Y_i \mid \text{NoFocal}] + (1-p) \cdot \mathbb{E}[Y_i \mid \text{NoRandom}] \tag{B.3}$$

$$\mathbb{E}[Y_i \mid \text{NoFocal}] = \mathbb{E}[Y_i \mid \text{NoRandom}] - 1/p \cdot (\mathbb{E}[Y_i \mid \text{NoRandom}] - \mathbb{E}[Y_i|\text{NonFocal}])$$

$$\mathbb{E}[Y_i(0)] = \bar{Y}(\text{NoRandom}) - 1/p \cdot (\bar{Y}(\text{NoRandom}) - \bar{Y}(\text{NonFocal})) \tag{B.4}$$

We obtain Equation (B.1) by combining Equations (B.2) and (B.4). □

Variance

Under the SUTVA assumption, the observations in groups Focal_b, NonFocal_b, and NoRandom_b are independent, so the variance of the proposed estimator is

$$\mathbb{V}(\text{ATE}) = \mathbb{V}\left(\bar{Y}(\text{Focal})\right) + \left(\frac{p-1}{p}\right)^2 \mathbb{V}\left(\bar{Y}(\text{NoRandom})\right) + \left(\frac{1}{p}\right)^2 \mathbb{V}\left(\bar{Y}(\text{NonFocal})\right). \tag{B.5}$$

Illustrative Example

The second term in Equation (B.1) captures the average effect of targeted focal coupons across the entire customer population. It may seem counterintuitive that this term does not depend on the share of targeted focal coupons or the targeting rules themselves. We provide an illustrative example to clarify the underlying intuition for the estimator.

Let λ equal the proportion of targeted focal coupons in the *NoRandom* group, t equal the average treatment effect of a targeted coupon on the targeted population, and p equal a probability that a random coupon replaces a specific targeted coupon on a printout. We notice that t is generally not equivalent to ATE, because targeting rules introduce selection. The average impact of focal targeted coupons across the entire population is λt. We want to demonstrate that we can estimate λt using the difference in (per customer) outcomes between the *NoRandom* and *NonFocal* groups as:

$$\lambda t = \frac{1}{p}\left(\bar{Y}(\text{No Random}) - \bar{Y}(\text{NonFocal})\right) \tag{B.6}$$

Imagine 10,000 customers receive coupons. The platform randomly assigns 500 customers to receive random non-focal coupons, and the targeting algorithm assigns 1,000 customers to receive targeted focal coupons, meaning λ=10%. The remaining coupons received by these customers are targeted non-focal coupons; these coupons have no impact on purchases of the focal brand, and are irrelevant for our example. In our identification approach, these 10,000 customers constitute the *NoRandom* and *NonFocal* groups.

Because of the random assignment, 50 (10%) of the 500 non-focal random coupons are assigned to the same customers as the targeted focal coupons. Among these 50 customers, 1 out of 7 (i.e., 7.1 customers; $p = 1/7$) of the targeted focal coupons are replaced by the random non-focal coupon. This results in the following *NoRandom* and *NonFocal* groups:

	NoRandom	**NonFocal**
Total customers	9,500	500
Targeted focal coupon:		
Yes	950 (10%)	42.9 (8.6%)
No	8,550 (90%)	457.1 (91.4%)

Notes: We are rounding $\lambda \cdot (1-p) \cdot 100\% = 0.1 \cdot \frac{6}{7} \cdot 100\% \approx 8.6\%$.

Let b indicate the baseline level of spending per customer without a coupon for the focal brand. The *NoRandom* average per customer outcome is: $b + \lambda t = b + 0.1t$ units, and the *NonFocal* average per customer outcome is: $b + (1-p)\lambda t = b + 0.086t$ units. This yields the following equivalence:

$$\frac{1}{p}\left[\bar{Y}(\text{NoRandom}) - \bar{Y}(\text{NonFocal})\right] = \frac{1}{p}\left[(b + 0.1t) - (b + 0.086t)\right] = \frac{1}{p}\left[\lambda t p\right] = \lambda t \tag{B.7}$$

This example illustrates how our adjustment to the identification in Equation 2 correctly accounts for the treatment effect introduced by targeted coupons. Our identification approach does not require the share of customers targeted by the focal coupons or the average treatment effects on the targeted. It builds on the average outcomes in the *NoRandom* and *NonFocal* conditions, weighted by how frequently targeted coupons are replaced by random ones in the *NonFocal* group.

Appendix C. Category Expansion and Stockpiling Measures

In this appendix, we examine how category-level ATEs (Immediate and Post-Period) vary with category-level stockpiling characteristics: *Easy*, *Like*, and the composite *Stockpiling* score (see Web Appendix F.1). Table C.1 reveals that none of the correlations are statistically significant ($p > 0.10$).

Table C.1: ATE Correlations with Category Stockpiling Scores

(A) Category Purchasers

	Easy	**Like**	**Stockpiling**
Immediate ATE	−0.067	−0.093	−0.088
	(0.580)	(0.444)	(0.469)
Post-Period ATE	−0.010	−0.052	−0.038
	(0.934)	(0.668)	(0.758)

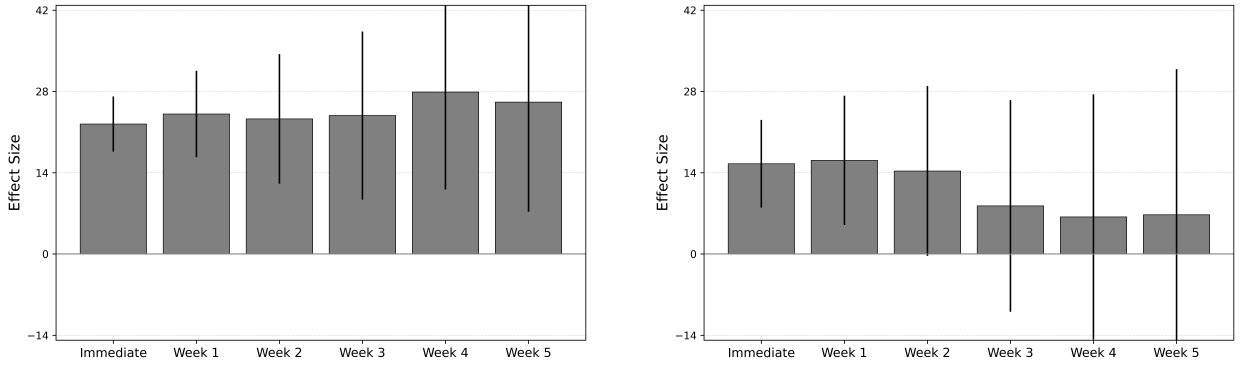
(B) All Respondents

	Easy	**Like**	**Stockpiling**
Immediate ATE	−0.056	−0.093	−0.082
	(0.644)	(0.445)	(0.500)
Post-Period ATE	−0.059	−0.080	−0.076
	(0.625)	(0.508)	(0.533)

Notes: The table reports the pairwise correlation between each ATE and each category-level characteristic. The sample size for each correlation is 71 (categories), and p-values are reported in parentheses.

In Figure C.1, we compare *Quantity* ATEs between urban and suburban stores. If stockpiling were a major driver of coupon effectiveness, we would expect suburban locations to exhibit larger immediate sales increases and deeper post-promotion dips since these customers have more room to stockpile. Instead, if anything, we observe larger immediate effects in urban locations. There is an indication of a post-promotion dip in suburban stores, but it is not statistically significant, and does not fully offset the initial gains. In urban stores, it is clear that coupon effectiveness extends beyond any stockpiling effects.

Figure C.1: Quantity ATEs in Urban and Suburban Stores



(A) Urban Stores

(B) Suburban Stores

Notes: The figures report category-level *Quantity* ATEs averaged across the 71 categories. In Panel (A), we calculate ATEs using purchases at the focal retailer's stores located in urban areas. In Panel (B), we calculate ATEs using purchases at stores located in suburban areas. Error bars indicate 95% confidence intervals.

Web Appendix to

Store-Entry Coupons: A Large-Scale Field Experiment

August 2026

A Category-Level Average Treatment Effects **WA-2**

B Randomization Checks **WA-5**

C Robustness to SUTVA and Competitive Coupons **WA-6**

D Summary Statistics for Brand Features **WA-10**

E Additional Results for Discount CATEs **WA-12**

F Stockpiling and Impulse **WA-14**

G Subsequent Coupon Delivery and Algorithm Feedback **WA-17**

H Panel Data: Difference-in-Differences **WA-19**

I Panel Data: Evidence of Multi-Homing **WA-23**

J Near vs. Far From Competitors: Urban vs. Suburban **WA-25**

Web Appendix A Category-Level Average Treatment Effects

Table WA-A.1: Category-Level Average Treatment Effects

Group	Category	Incidence ATE	Quantity ATE	Revenue ATE
Drinks	**Drinks Alcoholic**			
	Rum blend	0.66%	12.86	€119.16
	Scotch whisky	1.02%	9.52	€82.29
	Spirit-based mixed drinks (premixes)	0.33%	6.06	€14.15
	Drinks Non Alcoholic			
	Carbonated soft drinks (soda)	1.16%	10.45	€11.19
	Energy drinks	–0.07%	8.93	–€2.36
	Fruit juices / sweet musts	1.73%	30.48	€27.17
	Iced tea / iced coffee	–0.02%	9.17	–€6.53
	Sparkling mineral water	–1.21%	–34.19	–€80.98
Food	**Bread Cake**			
	Packaged ready-made cakes	–0.35%	-3.94	–€10.77
	Packaged rusk (zwieback)	0.83%	9.96	€6.52
	Condiments			
	Cooking oil (not refrigerated)	1.50%	24.10	€56.18
	Gourmet sauces	0.68%	0.10	€23.26
	Seasonings (MSG etc.)	0.95%	40.92	€41.98
	Convenience			
	Ready meals [P]	1.80%	54.72	€45.14
	Soups, liquid	1.82%	52.75	€66.74
	Dairy			
	Cheese (self-service) [P]	3.06%	46.42	€50.02
	Dairy products (misc.) [P]	1.02%	33.45	–€0.66
	Fresh milk [P]	1.30%	54.76	€49.68
	Milk-based mixed drinks	2.12%	38.60	€105.61
	Quark / curd cheese [P]	2.01%	40.99	€23.60
	Quark desserts & preparations [P]	3.54%	33.89	€35.77
	UHT milk (long-life milk)	–3.33%	–41.08	–€25.40
	Yogurt (incl. preparations) [P]	0.96%	10.58	€23.44

Continued on next page

Table WA-A.1 – continued from previous page

Group	Category	Incidence ATE	Quantity ATE	Revenue ATE
	Frozen Food			
	Fish fillets (plain)	1.29%	22.16	€93.48
	Leaf & stem vegetables	1.64%	20.11	€29.80
	Pizzas & quiches	2.15%	40.41	€79.01
	Potato pancakes (latkes)	2.02%	22.29	€19.20
	Ready meals & casseroles	2.19%	37.65	€97.06
	Meat Sausages			
	Cooked sausage (sliced etc.) [P]	1.36%	13.78	€9.28
	Meat/ham sausage [P]	1.42%	15.49	€9.29
	Mortadella / bologna sausages [P]	1.41%	28.34	€27.67
	Raw sausage specialties (self-service) [P]	2.21%	32.42	€68.88
	Snacks			
	Chips & snack sticks	1.45%	32.55	€36.46
	Nuts, almonds & seeds/kernels	0.27%	7.34	€14.22
	Roasted peanuts	0.90%	18.73	€12.39
	Snack foods	1.50%	17.03	€19.24
	Staples			
	Baking mixes (household)	0.88%	19.59	€18.08
	Condensed milk	1.42%	34.89	€17.53
	Honey	2.06%	29.18	€62.25
	Jams / preserves	0.69%	–1.91	€5.12
	Long-grain rice	0.85%	10.27	€10.79
	Muesli / granola	0.77%	14.39	€34.08
	Nut & chocolate spreads	2.33%	25.76	€39.99
	Oat flakes / cereal flakes	1.67%	26.78	€24.79
	Pasta (e.g., glass noodles)	1.05%	28.16	€61.37
	Processed rice	1.68%	44.27	€63.62
	Sweets			
	Alcohol chocolates	0.74%	16.33	€32.91
	Butter biscuits	0.43%	6.58	–€3.29
	Chocolate bars (incl. candy bars)	1.48%	16.93	€29.55

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Table WA-A.1 – continued from previous page

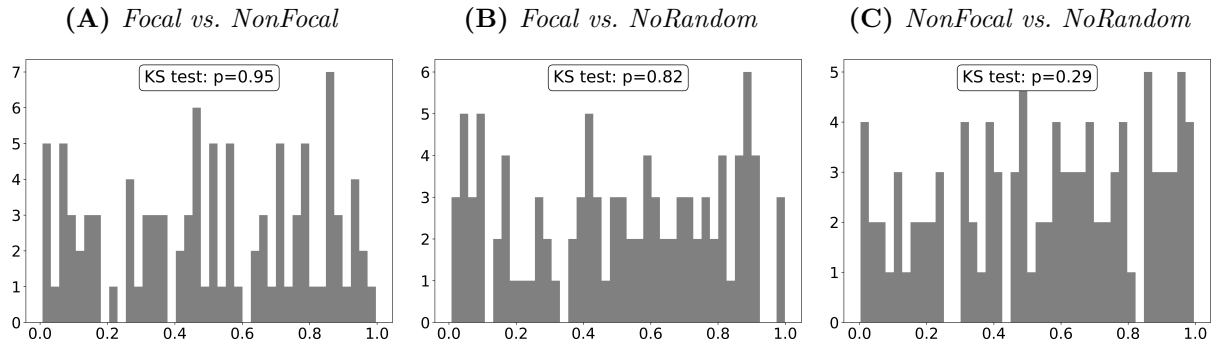
Group	Category	Incidence ATE	Quantity ATE	Revenue ATE
	Chocolate biscuits/pastries	1.96%	27.03	€30.84
	Chocolate products	2.61%	37.90	€51.97
	Filled chocolate tablets	2.01%	34.39	€44.12
	Sandwich cookies	–0.02%	–1.14	€0.09
	Single chocolates (assorted)	1.04%	16.14	€45.14
	Solid chocolate tablets	1.71%	36.39	€25.37
	Sugar confectionery	1.32%	34.81	€20.40
	Tea Coffee			
	Ground roasted coffee	–0.22%	–2.07	–€33.40
Near Food	**Body Care**			
	Deodorants (aerosol etc.) [F]	1.29%	18.91	€23.69
	Facial tissues [F]	–0.55%	–11.05	–€18.59
	Hair sprays & hair lacquer [F]	0.80%	8.27	€16.31
	Kitchen paper towels [F]	1.42%	11.27	€13.75
	Shampoo / hair wash products [F]	0.89%	10.45	€14.47
	Shower gel / body wash [F]	1.50%	19.15	€25.51
	Toilet paper [F]	1.96%	23.63	€66.03
	Toothpaste [F]	0.67%	9.26	€14.40
	Home Care			
	Dishwasher detergent (machine) [F]	0.48%	4.44	€16.25
	Laundry Care			
	Fabric softener / laundry rinse [F]	2.11%	27.89	€44.30
	Liquid laundry detergent [F]	0.08%	–0.28	€1.49
Pet Care	**Pet Food**			
	Dog snacks & treats	–0.01%	1.88	€1.42
	Wet cat food [F]	1.08%	61.73	€54.15
	Wet dog food [F]	0.29%	12.25	€11.17

Notes: The table reports the estimated immediate treatment effects for all product categories in our study. The index [P] denotes perishable product categories, and the index [F] refers to product categories with fixed (inelastic) consumption. The *Quantity* and *Revenue* ATEs are reported per 1,000 coupons (per mille).

Web Appendix B Randomization Checks

The randomization in our experiment is implemented programmatically in the system and was separately verified with the engineering team. Reassuringly, campaigns in our sample pass the randomization tests. We used an F-test with three pretreatment variables to test randomization. The pretreatment variables include the total spending, recency, and average basket size in 90 days prior to obtaining a coupon (specific to the customer×date observation). These variables are different from the ones used to detect eligibility criteria in Appendix A. We report the distribution of p-values across the campaigns for *Focal vs. NonFocal*, *Focal vs. NoRandom*, and *NonFocal vs. NoRandom* in Figure WA-B.1. The distributions are not statistically different from the uniform distribution using the Kolmogorov-Smirnov test. For completeness, we also implemented a randomization check on the discount levels (recall that the experiment randomized both brand assignments and discount levels). The randomization check for the discounts revealed no concerns with the implementation of the randomization.

Figure WA-B.1: Distribution of p-values in the Randomization Tests

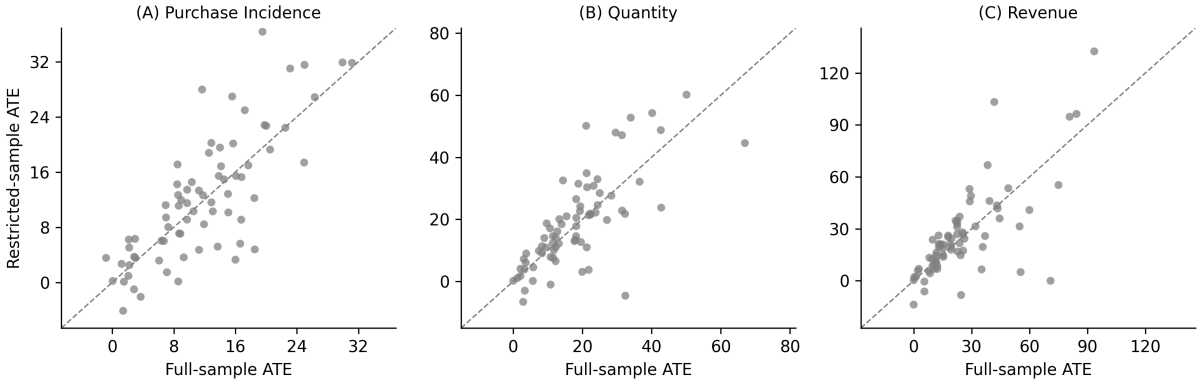


The figure reports histograms of the distribution of p-values in randomization tests. The sample size in each histogram is 101 (campaigns). The y-axis is a count of the number of brands, and the x-axis is p-values in the F-test with three pretreatment variables.

Web Appendix C Robustness to SUTVA and Competitive Coupons

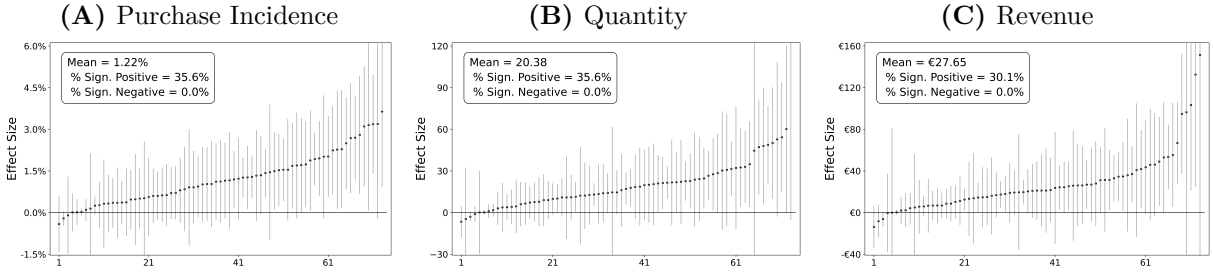
We replicate our main findings using a reduced data sample that focuses on the first observation per customer (for each campaign), and only includes campaigns that had no competitive coupons during the observation period. The reduced sample mitigates potential SUTVA violations and competitive spillover effects. The analysis in this section uses a substantially smaller sample: 26,478 random coupons across 73 campaigns, compared to 88,840 random coupons across 101 campaigns in the main text. However, the pattern of results is consistent with the key findings summarized in Table 4.

Figure WA-C.1: Campaign-Level ATEs: Restricted vs. Full Sample



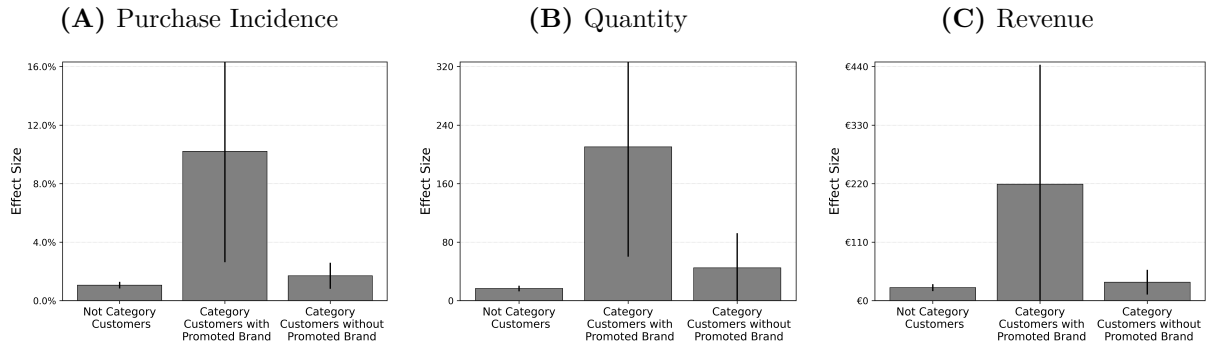
Notes: The figure compares campaign-level immediate ATEs in the restricted and full samples for the 73 campaigns present in both samples (45-degree line shown). The campaign-level estimates correlate at approximately 0.80 for all three outcome measures, and the paired differences are insignificant ($p \geq .33$). Applying each campaign's observed standard-error inflation to its full-sample t-statistic predicts that 22–32% of the campaign-level effects would remain significant at the 5% level (across the three outcome measures), which accounts for most of the observed 30–36%.

Figure WA-C.2: Distribution of Average Treatment Effects (ATEs)



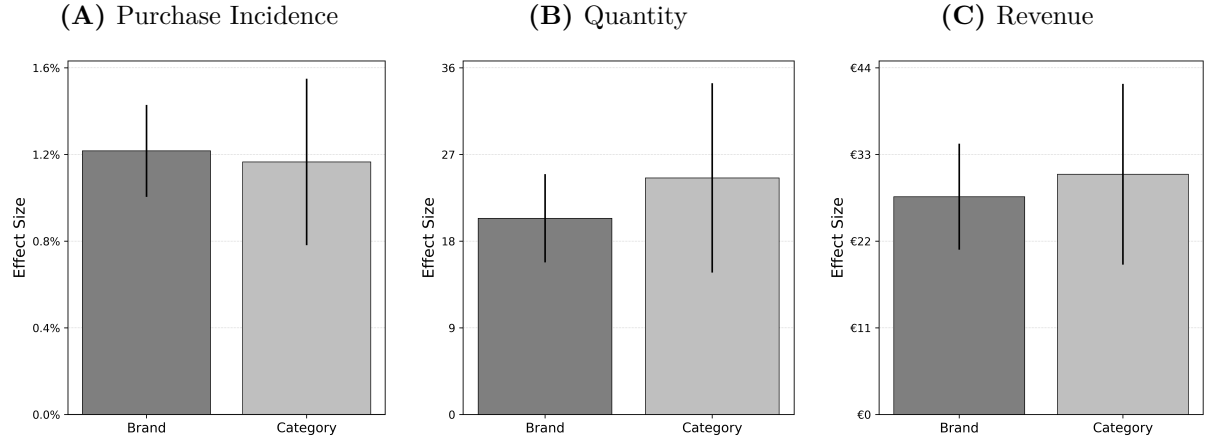
Notes: The figure summarizes the distribution of average treatment effects for the 73 brands that faced no competitive coupons during the experiment. Panel A reports the incremental purchase incidence for each brand (in percentage points). Panel B reports the incremental quantity sold (in units per 1,000 coupons). Panel C reports the incremental revenue (in €per 1,000 coupons). Error bars indicate 95% confidence intervals.

Figure WA-C.3: Heterogeneity in Treatment Effects for the Promoted Brand



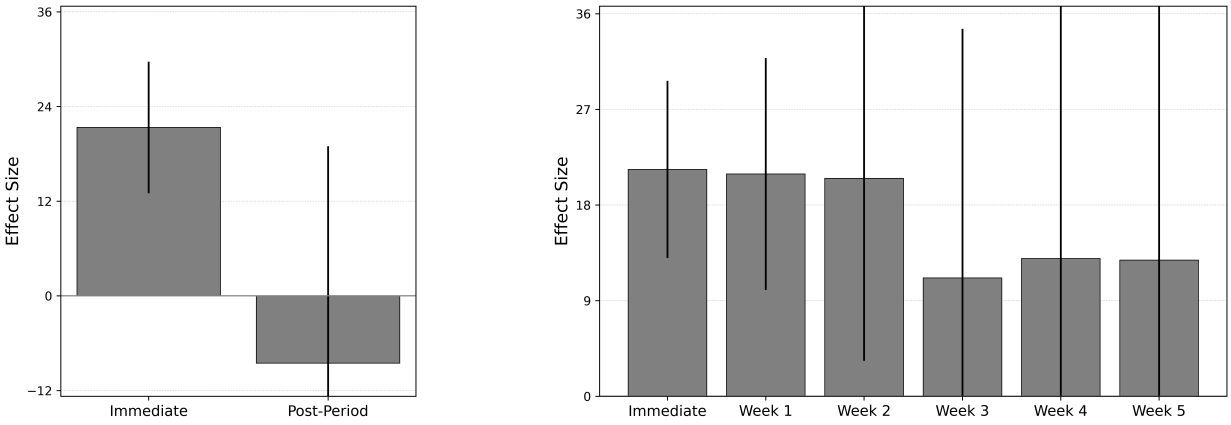
Notes: The figures report the average ATEs (on the current trip) averaged across the 73 focal campaigns that faced no competitive coupons. Each column represents a different customer segment. Error bars indicate 95% confidence intervals.

Figure WA-C.4: Comparison of Brand and Category ATEs



Notes: The figures report the average Brand and Category ATEs (on the current trip) averaged across the 73 focal campaigns that faced no competitive coupons. Error bars indicate 95% confidence intervals.

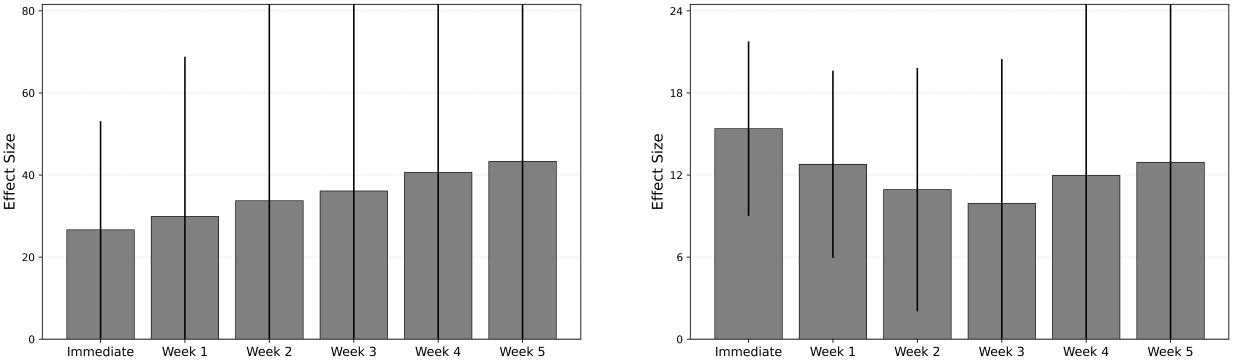
Figure WA-C.5: Temporal Substitution: Immediate and Post-Period Category-Level ATEs



(A) Distinct Periods **(B)** Cumulative Periods

Notes: These figures plot category-level *Quantity* ATEs averaged across the 61 categories in which only one brand distributed coupons (no competitive coupons). In Panel (A), we report separate findings for the Immediate and Post-Period. In Panel (B), we report cumulative findings. The negative Post-Period point estimate may invite a stockpiling interpretation, and so we examined it formally. However, further investigation confirms that the dip is not statistically significantly different from zero or different from the outcome for the unrestricted sample. Error bars indicate 95% confidence intervals.

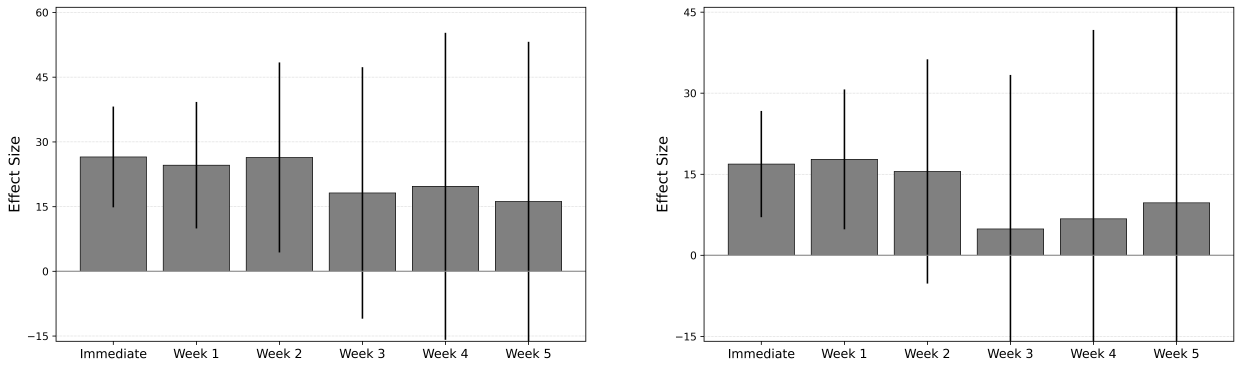
Figure WA-C.6: Categories in Which We Do Not Anticipate Changes in Consumption



(A) All Customers **(B)** Customers Without a Category Purchase

Notes: The figures report cumulative category-level *Quantity* ATEs averaged across categories in which we do not expect coupons to cause changes in consumption. Panel (A) includes all customers, Panel (B) includes customers without a prior category purchase at the focal retailer in the 90 days before customers received the store-entry coupon. Error bars indicate 95% confidence intervals.

Figure WA-C.7: Competing Stores Close or Far From Focal Retailer's Stores



(A) Competing Stores Close **(B)** Competing Stores Far

Notes: The figures report category-level *Quantity* ATEs averaged across the 61 categories where only one brand distributed coupons (no competitive coupons). In Panel (A) we calculate ATEs using purchases at the focal retailer's stores located close to competing stores. In Panel (B) we calculate ATEs using purchases at stores located far from competing stores. Error bars indicate 95% confidence intervals.

Web Appendix D Summary Statistics for Brand Features

This appendix provides descriptive statistics and correlation patterns for the brand-level variables used in the main text. Table WA-D.1 reports summary statistics for four brand features that have appeared prominently in the price promotions literature (Narasimhan et al. 1996; Bell et al. 1999; Ailawadi and Neslin 1998):

Brand Penetration: Percentage of customers who purchased at least one item from the brand in the data period.

Brand Loyalty: Spending on the brand as a % of total spending in the category. Calculated first at the customer level (among customers of the focal brand), and then averaged across customers.

Price Position: Average price paid for the brand, divided by average price paid in the category.

Interpurchase Time: Average (across customers) in the number of days between each customer's sequential purchases of the brand.

These features are all constructed using the retailer's transaction data in the three months prior to the store-entry coupon experiments.

Table WA-D.1: Brand Characteristics

Variable	Average	Std. Dev	$P_{25\%}$	$P_{75\%}$
Brand Penetration	1.27%	1.24%	0.49%	1.47%
Brand Loyalty	0.76	0.14	0.66	0.87
Price Position	1.17	0.64	0.84	1.38
Interpurchase Time	18.53	5.32	15.43	21.13

Notes: *Brand Penetration*, *Brand Loyalty*, *Price Position*, and *Interpurchase Time* are based on the retailer's loyalty card data in the three months prior to the store-entry coupon experiments.

In Table WA-D.2, we report the pairwise correlation between the three treatment effects (incidence, quantity, and revenue) and four brand features. *Brand Penetration* explains more variation in store-entry coupon treatment effects than brand loyalty, price position, or interpurchase time. Brands that are purchased by more customers tend to have larger ATEs. One explanation for the effect is that some categories are not relevant for certain customers, and coupons do not change this relevance. If few customers have pets, then we would expect both a low penetration rate for pet food brands, and a relatively small response to pet food coupons.

Quantity treatment effects are larger for brands with less loyal customers. A possible interpretation is that when *Brand Loyalty* is low, customers in the focal category are more price-sensitive or have lower switching costs, so they frequently switch brands during a promotion. For these customers, the coupon discount is sufficient to increase preferences for the brand and prompt a purchase.

Table WA-D.2: Pairwise Correlation Between Treatment Effects and Brand Characteristics

	Incidence ATE	**Quantity ATE**	**Revenue ATE**
Brand Penetration	0.41 **	0.46 **	0.20 **
Brand Loyalty	–0.13	–0.28 **	0.12
Price Position	–0.13	–0.18 †	0.04
Interpurchase Time	0.07	–0.09	0.12

The table reports the Spearman pairwise correlations between each brand feature and purchase incidence, purchase quantity, and revenue treatment effects (on the current trip). The unit of analysis is a brand, and the sample size is 101. ** indicates significantly different from zero, $p < 0.01$. * indicates significantly different from zero, $p < 0.05$. † indicates significantly different from zero, $p < 0.10$.

Web Appendix E Additional Results for Discount CATEs

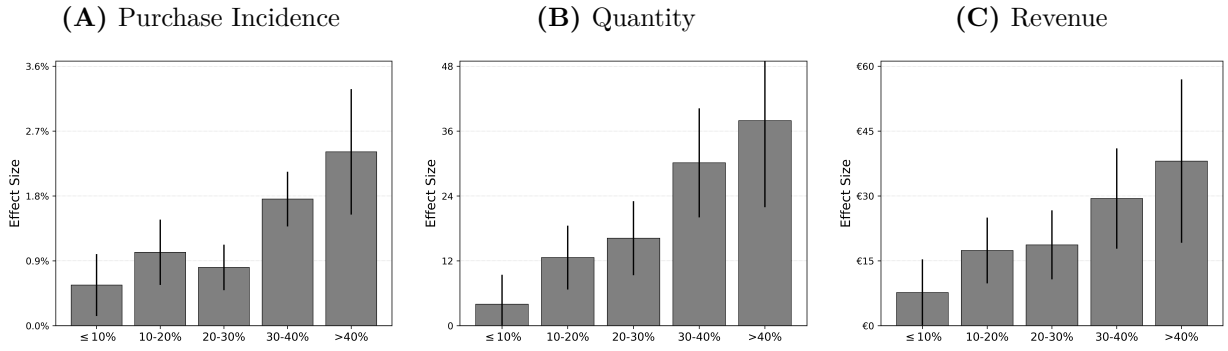
E.1 Discount CATEs Holding the Sample of Brands Fixed

In Section 5.3, we study how coupon treatment effects vary with discount depth by binning coupons into five 10-percentage-point discount buckets and estimating treatment effects by brand and discount bin, then averaging the brand-level treatment effects within each bucket. Because manufacturers do not necessarily offer the full range of discount levels for every brand, some brands have no observations in the lowest and/or highest discount bins. As a result, the set of brands contributing to the averages in Figure 5 can differ across discount bins. In this appendix, we ensure that comparisons across discount levels are not influenced by changes in the set of brands included in bins. We replicate the discount-depth analysis on a balanced set of 24 brands that have observations in all five discount buckets.

We follow the same identification strategy as in the main analysis. Treatment effects are first computed at the brand-by-discount-bin level and then averaged across brands within each discount bin. Figure WA-E.1 reports these averages, with error bars reflecting 95% confidence intervals.

The results in Figure WA-E.1 mirror the results in Figure 5 in the main text. Purchase incidence and quantity treatment effects are positive across all discount bins and generally increase with discount depth. For revenue, the balanced-sample estimates remain positive and increase with discount depth. The only contrast is that in Figure 5 the revenue effect dips in the last discount bin, while in Figure WA-E.1 the results for the fixed sample do not show a clear drop in the revenue effect for discounts above 40%. Overall, restricting the analysis to brands with support in all discount bins leaves the qualitative conclusion unchanged: deeper discounts generate larger treatment effects, and the main heterogeneity patterns are not driven by differences in the brands represented in each discount bin.

Figure WA-E.1: Treatment Effects at Each Discount Level



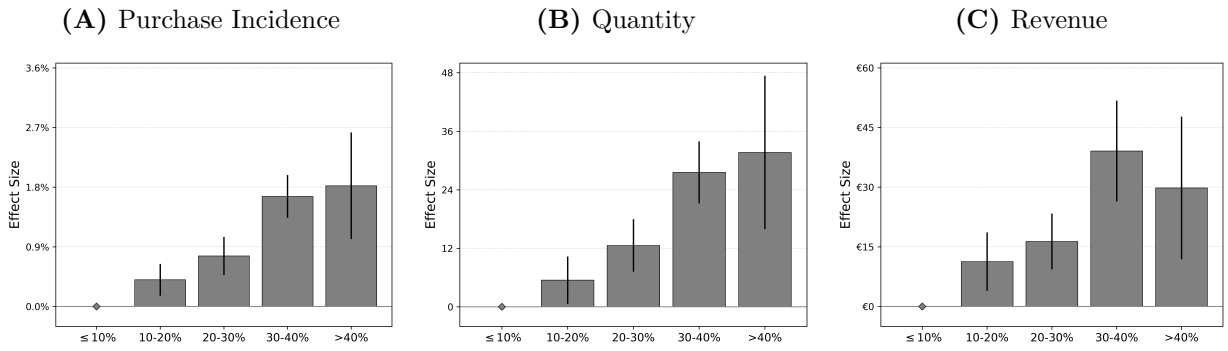
Notes: The figure depicts treatment effects as a function of the discount depth when restricting attention to the same set of 24 brands in each discount bin. Treatment effects within each discount bucket are first calculated at the brand level. The black dots represent means calculated by averaging the treatment effects across the brands. Error bars indicate 95% confidence intervals calculated using a non-parametric bootstrap.

E.2 Treated-Only Contrasts Across Discount Levels

Because discount depth is randomly assigned among coupon recipients, comparisons across discount levels in the *Focal* group are experimental and do not require a separate control group. This appendix reports how the three outcome variables (incidence, quantity, and revenue) varied with these contrasts in discount depth.

For each campaign, we bin the discounts into the same five bins as in Figure 5 of the main text (≤10%, 10–20%, 20–30%, 30–40%, and above 40%), and we use the lowest bin (≤10%) as the within-experiment baseline. We then compute the difference in Immediate brand-level outcomes between each bin and the campaign's own ≤10% bin (requiring at least 30 observations per bin). We report the average effects in Figure WA-E.2.

Figure WA-E.2: Treated-Only Contrasts Across Discount Levels



Notes: The figure reports immediate outcome differences relative to the ≤10% discount bin, with 95% confidence intervals calculated across campaigns. The diamond marks the ≤10% baseline (zero by construction).

The contrasts increase monotonically at every step for purchase incidence and quantity, and through the 30–40% bin for revenue. The >40% revenue estimate rests on only 24 campaigns and is statistically indistinguishable from the 30–40% bin. In a within-campaign linear specification, a 10-percentage-point increase in discount depth raises purchase incidence by 0.66% ($t = 10.0$), quantity by 11.30 (per 1,000 coupons; $t = 9.7$), and revenue by €14.42 (per 1,000 coupons, $t = 6.6$), with positive slopes in 84–89% of the 101 campaigns.

Web Appendix F Stockpiling and Impulse

F.1 Survey Design and Category-Level Scores

To investigate heterogeneity in category expansion, we use category-level measures capturing two behavioral mechanisms emphasized in the past literature: impulse purchasing and stockpiling. These measures are not derived from the retailer's sales data but are instead based on a consumer survey the retailer conducted in July 2017. The survey goal was to elicit category attributes reflecting well-documented differences in how consumers view and approach different product categories.

The stockpiling scores are based on the scale items developed by Narasimhan et al. (1996). Respondents rate each category on a 7-point Likert scale (1 = strongly disagree; 7 = strongly agree). The scale includes two items capturing perceived ease of storing extra quantities at home and the respondent's propensity to stock up when possible:

Easy "It is easy to store extra quantities of this product in my home."

Like "I like to stock up on this product when I can."

We construct a category-level measure of *Stockpiling* by averaging the responses to both of these questions across respondents.

The impulse scores are obtained using the two-item Impulse scale from Narasimhan et al. (1996):

Whim "I often buy this product on a whim when I pass by it in the store."

Urge "I typically like to buy this product when the urge strikes me."

Both questions are again measured using a 7-point Likert agreement scale. As with the *Stockpiling* measure, we average these two measures across respondents to create a measure of how susceptible a category is to being purchased on impulse.

The retailer recruited a panel of German consumers on Amazon Mechanical Turk (MTurk). A total of 614 respondents participated in the survey, and each respondent evaluated eight product categories that were randomly selected from the full set of categories. Limiting the number of categories per respondent reduced respondent burden and mitigated potential fatigue or wearout effects. In addition to completing the category-rating scales, respondents were asked whether they had purchased from each category within the past six months. Both the order in which categories were presented and the order of the relevant question blocks were randomized to avoid order effects.

From the raw data, we construct category-level scores using (i) all available responses and (ii) the subset of responses from respondents who reported at least one category purchase in the previous six months. Table WA-F.1 reports descriptive statistics for the *Impulse* and *Stockpiling* scales. We note that the retailer conducted the survey using broader category definitions than those analyzed in our main text. Specifically, the retailer collected scores for 43 categories, whereas our study relies on 71. To evaluate the correlation between ATEs and these survey metrics, we map

the aggregate scores to our more granular product categories.

Across both samples, the stockpiling items (Easy and Like) exhibit the highest mean levels and relatively low dispersion, indicating broad agreement that many categories are easy to store and suitable for stocking up. Impulse items show more moderate means and greater heterogeneity, with Urge consistently rated higher than Whim. The distributions are broadly similar across the two samples, with slightly lower impulse ratings in Panel (B).

Table WA-F.1: Category-Level Survey Scores

(A) Category Purchasers

Item	Mean	SD	Percentiles				
			10%	25%	50%	75%	90%
Stockpiling							
Easy	5.328	0.567	4.525	4.896	5.245	5.790	6.166
Like	4.562	0.777	3.552	3.797	4.554	5.242	5.489
Impulse							
Urge	4.197	1.053	2.521	3.472	4.542	4.902	5.420
Whim	3.798	0.842	2.513	3.313	3.780	4.480	4.921

(B) All Respondents

Item	Mean	SD	Percentiles				
			10%	25%	50%	75%	90%
Stockpiling							
Easy	5.161	0.620	4.406	4.649	5.141	5.659	5.997
Like	4.287	0.813	3.317	3.625	4.269	4.896	5.393
Impulse							
Urge	3.993	0.992	2.493	3.300	4.085	4.634	5.166
Whim	3.573	0.803	2.454	3.203	3.453	4.191	4.479

F.2 Correlations between Impulse Scores and ATEs

In this appendix, we examine how category-level ATEs (Immediate and Post-Period) vary with category-level characteristics that are related to impulse buying: *Urge*, *Whim*, and *Impulse* (see Web Appendix F.1). Categories with higher impulse scores exhibit lower Post-Period treatment effects, but the relationship is not statistically significant.

Table WA-F.2: ATE Correlations with Category Impulse Scores

(A) Category Purchasers

	Whim	**Urge**	**Impulse**
Immediate ATE	0.187	0.136	0.161
	(0.120)	(0.262)	(0.184)
Post-Period ATE	−0.180	−0.199	−0.192
	(0.137)	(0.099)	(0.112)

(B) All Respondents

	Whim	**Urge**	**Impulse**
Immediate ATE	0.184	0.151	0.167
	(0.127)	(0.213)	(0.167)
Post-Period ATE	−0.193	−0.196	−0.197
	(0.109)	(0.103)	(0.103)

Notes: The table reports the pairwise correlation between each ATE and each category-level characteristic. The sample size for each correlation is 71 (categories), and p-values are reported in parentheses.

Web Appendix G Subsequent Coupon Delivery and Algorithm Feedback

In this section, we investigate whether exposure to a focal coupon in the experiment could trigger additional focal-brand coupons in later sessions. If the algorithm preferentially delivered focal-brand coupons to treated customers after the experimental session, the effects of these coupons could offset the post-promotion dip in customer behavior and explain the near-zero Post-Period effects in Figure 8.

Table WA-G.1 evaluates whether subsequent coupon delivery differs between customers who received a random coupon in the experiment (*Focal*) and customers who did not receive a random coupon for the focal brand (*NonFocal* and *NoRandom*). We focus on two outcome variables: the number of targeted focal-brand coupons a customer received in the 35 days following the experimental session (*Brand Coupons*), and the number of targeted coupons in the campaign's category in the same window (*Category Coupons*). For each campaign, we calculate the absolute standardized mean difference (SMD) between the *Focal* group and each of the two baseline conditions (*NonFocal* and *NoRandom*),

$$|\text{SMD}| = \frac{|\,\bar{y}_{\text{Focal}} - \bar{y}_{\text{Baseline}}\,|}{\sqrt{\text{var}_{\text{Focal}} + \text{var}_{\text{Baseline}}}}. \tag{WA-G.1}$$

We also report the distribution across campaigns in Figure WA-G.1.

Table WA-G.1: SMD in Subsequent Coupon Delivery

	Focal vs. NonFocal			**Focal vs. NoRandom**		
Outcome Variable	Mean	< 0.10	< 0.20	Mean	< 0.10	< 0.20
Brand Coupons	0.047	91%	100%	0.046	92%	100%
Category Coupons	0.041	97%	100%	0.041	96%	100%

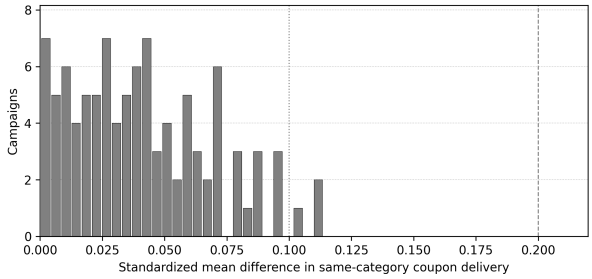
Notes: The table reports the mean absolute standardized mean difference in subsequent coupon delivery (days 1–35 after the experimental session) across the campaigns, together with the share of campaigns with standardized differences below 0.10 (the conventional negligibility threshold in balance diagnostics) and below 0.20 (the conventional threshold for a small effect).

The standardized differences are uniformly small. For brand coupons, the mean SMD is 0.047 against the *NonFocal* condition and 0.046 against the *NoRandom* condition; for category coupons, the mean is 0.041 in both comparisons. In both comparisons, at least 91% of campaigns fall below the conventional 0.10 negligibility threshold, and all campaigns fall below the 0.20 threshold for a small effect.[11] We conclude that it is unlikely that the treatment effects can be attributed to the targeting algorithm.

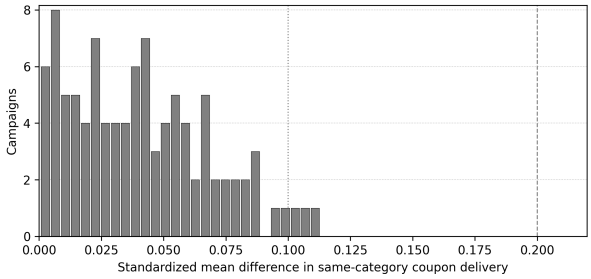
[11] Correcting the analysis for targeted focal coupons delivered during the experiment using Equation (2) leaves the conclusion unchanged.

Figure WA-G.1: SMD in Category Coupon Delivery

(A) Focal vs. NonFocal



(B) Focal vs. NoRandom



Notes: The figure reports the distribution of absolute standardized mean differences in subsequent same-category coupon delivery (days 1–35 after the experimental session), comparing the *Focal* condition with *NonFocal* or *NoRandom* condition. The dotted line indicates the conventional 0.10 negligibility threshold in balance diagnostics, and the dashed line represents the 0.20 threshold for a small effect.

Web Appendix H Panel Data: Difference-in-Differences

This appendix provides additional details on the difference-in-differences (DiD) analysis of household panel data that evaluates the impact of the store-entry coupon system on retailer sales (see Section 6).

H.1 Approach

We obtained household panel data covering purchases from grocery retailers for a sample of 46,150 German households. Panel participants upload receipts for all their grocery purchases. The data span a period of two years—one year before and one year after the coupon system introduction. For each panel household h, we know in which city the household lives at the time of the coupon system introduction.

Our coupon experiments, as well as the implementation of the coupon system, focus on a single metropolitan area. In contrast, the household panel covers all of Germany. Because the retailer also operates stores elsewhere in Germany without the coupon system, we can use a DiD approach to estimate the causal effect of the coupon system on household spending. We define a treatment indicator $Treated_r$ that equals 1 for households residing in the treated region and 0 for households residing in the untreated region. Furthermore, we define $Post_m$ as an indicator equal to 1 for months after the introduction of the coupon system and 0 for months before. In the DiD analysis, we focus on households with at least one purchase at the retailer before the coupon system introduction.

In each of the twelve months before and twelve months after the introduction of the coupon system, we separately aggregate spending at each retailer across households by month m and region r. We denote this aggregate sales as y_{rm}. We then estimate the following DiD specification separately for the focal retailer and for the main competitor:

$$\log y^c_{rm} = \tau^c \cdot (Treated_r \times Post_m) + \sigma_r + \sigma_m + \varepsilon_{rm} \tag{WA-H.1}$$

where c denotes either the focal retail chain or the competing chain, σ_r are region fixed effects, σ_m are month fixed effects, and ε_{rm} is a normally distributed error term.

The coefficient of interest is τ^c, which captures the change in spending for treated households at chain c after the introduction of the store-entry coupon system, relative to the corresponding change for control households. We report DiD estimates as percent changes and construct 95% confidence intervals using the regression standard errors. In addition to overall spending, we repeat the analysis for the nine product groups shown in the main text. Table WA-H.1 summarizes the results.

H.2 Balance Check

A potential concern in the panel-data DiD design is that we did not randomly assign households to treatment and control. Consequently, households in the treated region could differ systematically

Table WA-H.1: Difference-in-Differences Estimates

	Focal Retailer			Main Competitor		
	$\hat{\tau}$	SE	p-value	$\hat{\tau}$	SE	p-value
All	8.07%	2.74%	0.007	-3.88%	2.10%	0.077
Body & personal care	9.90%	8.66%	0.265	−1.06%	8.73%	0.905
Bread	4.63%	5.62%	0.419	−2.61%	3.56%	0.471
Candy & snacks	17.32%	4.55%	0.001	−3.55%	3.42%	0.309
Dairy	2.63%	3.30%	0.433	−3.14%	2.81%	0.276
Drinks	−0.92%	3.46%	0.793	−7.85%	4.48%	0.093
Food staples	7.91%	2.92%	0.013	−5.71%	2.57%	0.036
Fresh produce	8.94%	3.94%	0.033	0.04%	3.61%	0.992
Home & laundry care	15.55%	8.09%	0.067	−5.28%	8.03%	0.517
Meat & fish	12.88%	4.55%	0.009	−5.94%	3.58%	0.110

Notes: The table reports coefficients and standard errors from estimating Equation WA-H.1. Each treatment effect $\hat{\tau}$ is estimated in a separate regression. The unit of observation is region×month, and the sample size in each model is 48.

from households in the untreated region. We implement a balance check by estimating a logit model where the dependent variable equals one if the household resides in the treated region (and zero otherwise). The explanatory variables are measured before the introduction of the coupon system and include both pre-period purchasing behavior and household demographics: the household's mean and standard deviation of spending (in euros), the Herfindahl index of households' spending concentration across retailers, and demographic indicators (age groups, household size groups, and social classes).

Table WA-H.2 reports the estimated coefficients. Consistent with random differences in observables, none of the pre-treatment covariates is statistically significant at conventional levels. Overall, this balance check suggests that treated vs. control status is not predicted by observable pre-period purchasing patterns or demographic characteristics. This supports the comparability of the treated and control samples.

H.3 Placebo Test Results

A central identifying assumption in our panel-data DiD design is that, absent the introduction of the store-entry coupon system, outcomes for households in the treated region would have evolved in parallel to outcomes for households in the control region. A standard diagnostic is a *placebo timing test*: we assign a *fake* introduction date in a period when the coupon system was not yet operating and re-run the DiD. In this test, we should not find systematic treatment effects.

We implement the placebo by replicating the main DiD specification but shifting the post indicator to a pre-treatment month. Specifically, we keep the treatment definition unchanged, focus on purchases at the focal retailer, and aggregate purchases to the month×treatment-status level (overall and separately for each of the nine product groups used in the panel data). We then define

Table WA-H.2: Balance Check Results

Variable	Coefficient	Std. Err.	p-value	95% CI
Intercept	0.3612	0.222	0.104	[−0.074, 0.797]
Spending average	−0.0315	0.153	0.837	[−0.331, 0.268]
Spending variability	−0.1326	0.099	0.179	[−0.326, 0.061]
Spending concentration	0.0815	0.234	0.727	[−0.376, 0.539]
Age low	0.2414	0.263	0.360	[−0.275, 0.758]
Age high	0.0829	0.114	0.468	[−0.141, 0.307]
Large households	0.1164	0.320	0.716	[−0.510, 0.743]
Low social class	−0.0943	0.134	0.481	[−0.356, 0.168]
High social class	−0.0333	0.129	0.796	[−0.286, 0.220]

Notes: The table reports coefficients and standard errors from estimating a logit model where the dependent variable equals one if the household resides in the treated region (and zero otherwise). The unit of analysis is a household, and the sample size is 1,862.

a placebo post indicator that is 0 in the first six months and 1 in the last six months of the original time window (i.e., before the actual introduction of the couponing system). We then estimate the same DiD regression as in the main analysis.

The results in Table WA-H.3 show that the placebo estimates provide no evidence of an effect around the fake introduction date. The overall placebo DiD coefficient is small (0.67%) and statistically insignificant ($p = 0.87$). At the product-group level, placebo coefficients fluctuate in sign and magnitude but are uniformly imprecise and not statistically significant at conventional levels (all p-values exceed 0.10). The absence of systematic or statistically significant placebo effects supports the interpretation that the main DiD results are driven by the actual introduction of the coupon system rather than differential pre-trends or unrelated shocks in the treated region.

Table WA-H.3: Placebo Difference-in-Differences Estimates

	$\hat{\tau}$	Std. Err.	p-value
All	0.67%	3.99%	0.871
Body & personal care	−15.07%	8.72%	0.118
Bread	12.95%	8.32%	0.154
Candy & snacks	5.05%	6.56%	0.461
Dairy	−4.62%	4.64%	0.345
Drinks	3.34%	4.34%	0.461
Food staples	1.96%	4.84%	0.695
Fresh produce	3.06%	6.56%	0.653
Home & laundry care	0.14%	12.87%	0.991
Meat & fish	−3.26%	4.62%	0.497

Notes: The table reports coefficients and standard errors from estimating Equation WA-H.1 using a placebo treatment date. Each treatment effect $\hat{\tau}$ is estimated in a separate regression. The unit of observation is region×month, and the sample size is 24.

Web Appendix I Panel Data: Evidence of Multi-Homing

The proposed store-switching explanation, that customers purchase the focal brand on the current trip instead of purchasing it at a competing retailer on a future trip, requires that customers visit multiple retailers. We use household panel data to investigate this premise in this appendix.

The household panel sample includes the households with at least one purchase at the focal chain and at least ten grocery shopping trips in each matched June–December window, comparing the year before the coupon system's introduction with the year of its introduction. Retail chains are counted across the full competitive set in the panel (grocery chains, drugstores, specialty food retail, and online), because these formats compete for the CPG categories that the coupons target. Share of wallet is the focal chain's share of the household's panel-recorded FMCG spend.

Multi-homing is pervasive: The median customer shops at 9 distinct retail chains. If we restrict attention to the top quartile of customers based upon spending at the focal retailer, 95% of households purchase at three or more competing chains. The median top-quartile household allocates about a fifth of its FMCG expenditure to the focal chain (see Table WA-I.1).

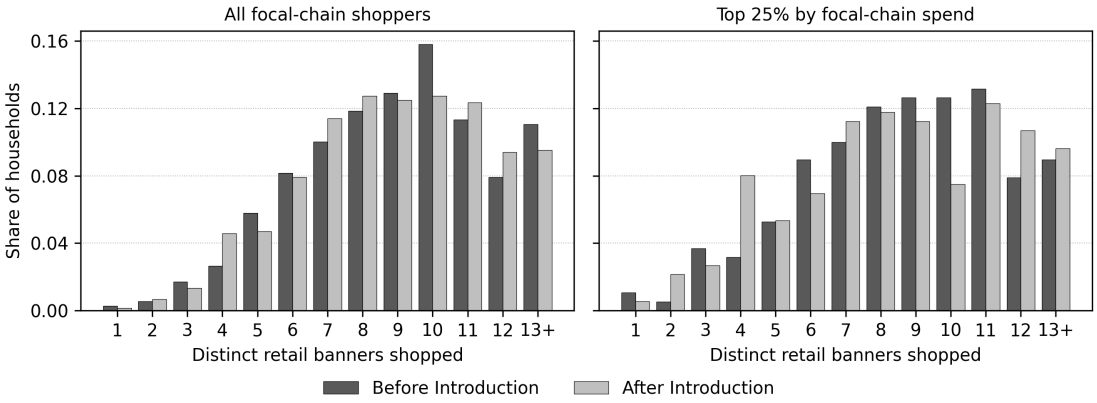
Table WA-I.1: Multi-Homing Among Households Shopping the Focal Chain

	All focal-chain shoppers		**Top quartile by focal spend**	
	Pre	Post	Pre	Post
Households	759	745	190	187
Distinct retail chains, mean (median)	9.0 (9)	9.0 (9)	8.7 (9)	8.6 (9)
Share shopping ≥2 competing chains	99.2%	99.2%	98.4%	97.3%
Share shopping ≥3 competing chains	97.5%	97.9%	94.7%	94.7%
Focal share of wallet, median (mean)	3.5% (9.4%)	3.0% (9.1%)	19.4% (26.1%)	19.3% (27.0%)
Share with focal share of wallet >50%	3.3%	3.2%	12.1%	12.8%

Notes: The table reports multi-homing statistics for panel households in the experimental region, in matched June–December windows before (Pre) and after (Post) the introduction of the coupon system. Restricting attention to grocery chains only yields medians of seven chains (all shoppers) and six chains (top quartile), with 92% and 85% of households shopping three or more competing chains.

We also investigated whether multi-homing changed after the introduction of the coupon system at the focal retailer. Figure WA-I.1 reveals that the distributions are statistically indistinguishable before and after the coupon system's introduction (Kolmogorov–Smirnov $p \geq .61$), and within households observed in both windows, chain counts at fixed shopping trips and focal expenditure shares are unchanged ($p = 0.18$ and $p = 0.13$). The coupon system's introduction neither reduced multi-homing nor concentrated expenditures.

Figure WA-I.1: Distinct Retail Chains Shopped



Web Appendix J Near vs. Far From Competitors: Urban vs. Suburban

This appendix repeats the near/far comparison of category treatment effects separately for urban and suburban stores. We classify the experimental stores by environment (89 urban and 50 suburban; 8 rural stores are dropped) and split each group at its median distance between the focal store and the nearest competitor. The first panel reports the same comparison for the full store sample, using the overall median split from Figure 11. We report the differences between the near and far stores using paired t-tests across 71 categories. Similar to the main text, we investigate the differences for all customers and for no-category customers (NCC), meaning customers without a purchase in the focal category at the focal retailer in the 90 days before the experiment. NCC rows are restricted to campaigns with at least 50 focal and 50 nonfocal observations in both store groups.

Table WA-J.1: Quantity ATEs for Near vs. Far Stores

Sample	Outcome	Near	Far	Diff (SEM)	p
Suburban stores					
All customers	Immediate	25.8	7.4	18.4 (7.3)	0.014
	Cumulative week 5	36.0	−20.1	56.0 (29.1)	0.059
NCC	Immediate	15.5	4.6	10.8 (5.0)	0.034
	Cumulative week 5	20.5	−9.3	29.8 (9.8)	0.003
Urban stores					
All customers	Immediate	21.2	23.7	−2.5 (4.5)	0.588
	Cumulative week 5	35.7	16.1	19.6 (29.5)	0.508
NCC	Immediate	15.7	19.6	−4.0 (3.6)	0.269
	Cumulative week 5	20.9	17.0	3.8 (10.5)	0.716

Notes: Near and Far report quantity ATEs (per mille) averaged over categories within each store group; Diff is the paired near − far difference with the standard error of the mean in parentheses. Immediate refers to the day of the coupon session (day 0) and cumulative week 5 to days 0–35. p-values are paired t-tests over 71 categories.